

3.5 Demand management (demand side policies) — monetary policy (includes HL only sub-topics)

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Level7 Education

Learning objectives

3.5 Demand management (demand side policies) – monetary policy (includes HL only sub-topics)	Depth	Diagrams and calculations
Monetary policy • Control of money supply and interest rates by the central bank	AO1	
Goals of monetary policy • Low and stable rate of inflation • Inflation targeting • Low unemployment • Reduce business cycle fluctuations • Promote a stable economic environment for long-term growth • External balance	AO2	

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Learning objectives

3.5 Demand management (demand side policies) – monetary policy (includes HL only sub-topics)	Depth	Diagrams and calculations
The process of money creation by commercial banks (HL only)	AO2	
Tools of monetary policy (HL only) • Open market operations • Minimum reserve requirements • Changes in the central bank minimum lending rate (base rate/discount rate/refinancing rate changes) • Quantitative easing	AO2	
Demand and supply of money – determination of equilibrium interest rates (HL only)	AO2 AO4	Diagram (HL only): showing the determination of equilibrium interest rates

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Learning objectives

3.5 Demand management (demand side policies) – monetary policy (includes HL only sub-topics)	Depth	Diagrams and calculations
Real versus nominal interest rates	AO2	Calculation: real interest rates from given data
Expansionary and contractionary monetary policies to close deflationary/recessionary and inflationary gaps	AO3 AO4	Diagram: AD/AS curves showing expansionary and contractionary monetary policy
Effectiveness of monetary policy • Constraints on monetary policy, including: • limited scope of reducing interest rates, when close to zero • low consumer and business confidence	AO3	

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Learning objectives

3.5 Demand management (demand side policies) – monetary policy (includes HL only sub-topics)	Depth	Diagrams and calculations
• Strengths of monetary policy, including: • incremental, flexible and easily reversible • short time lags • Strengths and limitations in promoting growth, low unemployment, and low and stable rate of inflation	AO3	

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Monetary policy

Monetary policy is carried out by the central bank and involves the control of money supply and interest rates to influence aggregate demand and subsequently fulfil macroeconomic objectives.

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Key terms

Central bank: The monetary authority responsible for an economy's monetary policy and financial system regulation.

Interest rates: The cost of borrowing and reward for saving of money, expressed as a percentage.

Money supply: The amount of money circulating in the economy, which includes notes and coins, loans, credits and deposits.

Demand-side policy: Government policy that aims to influence the level of AD of the economy.

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Goals of monetary policy

Using the sources below, compare and contrast the objectives of the three central banks:

Source 1: The Federal Reserve

Source 2: The Bank of England

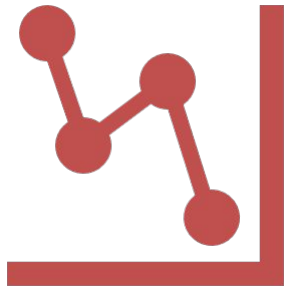
Source 3: European Central Bank

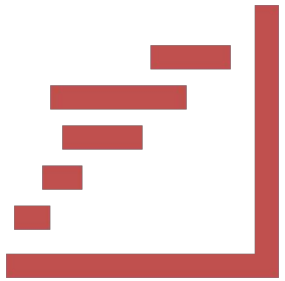
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Goals of monetary policy

Low and stable rate of inflation

An **inflation target** is the practice of using monetary policy to achieve a predetermined level of inflation. By increasing the transparency of the central bank and controlling inflation, this would create a stable economic environment for consumers and producers.

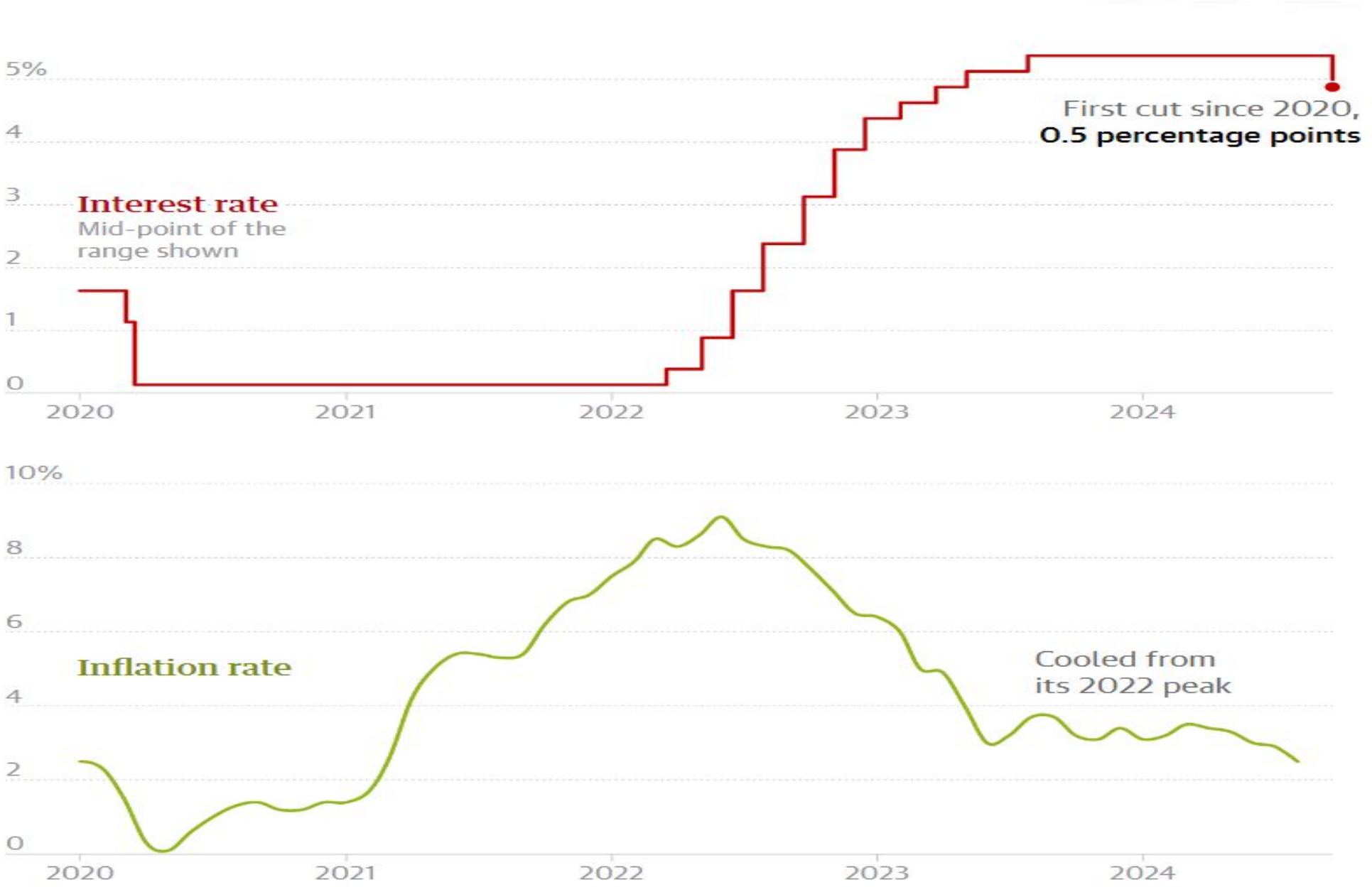






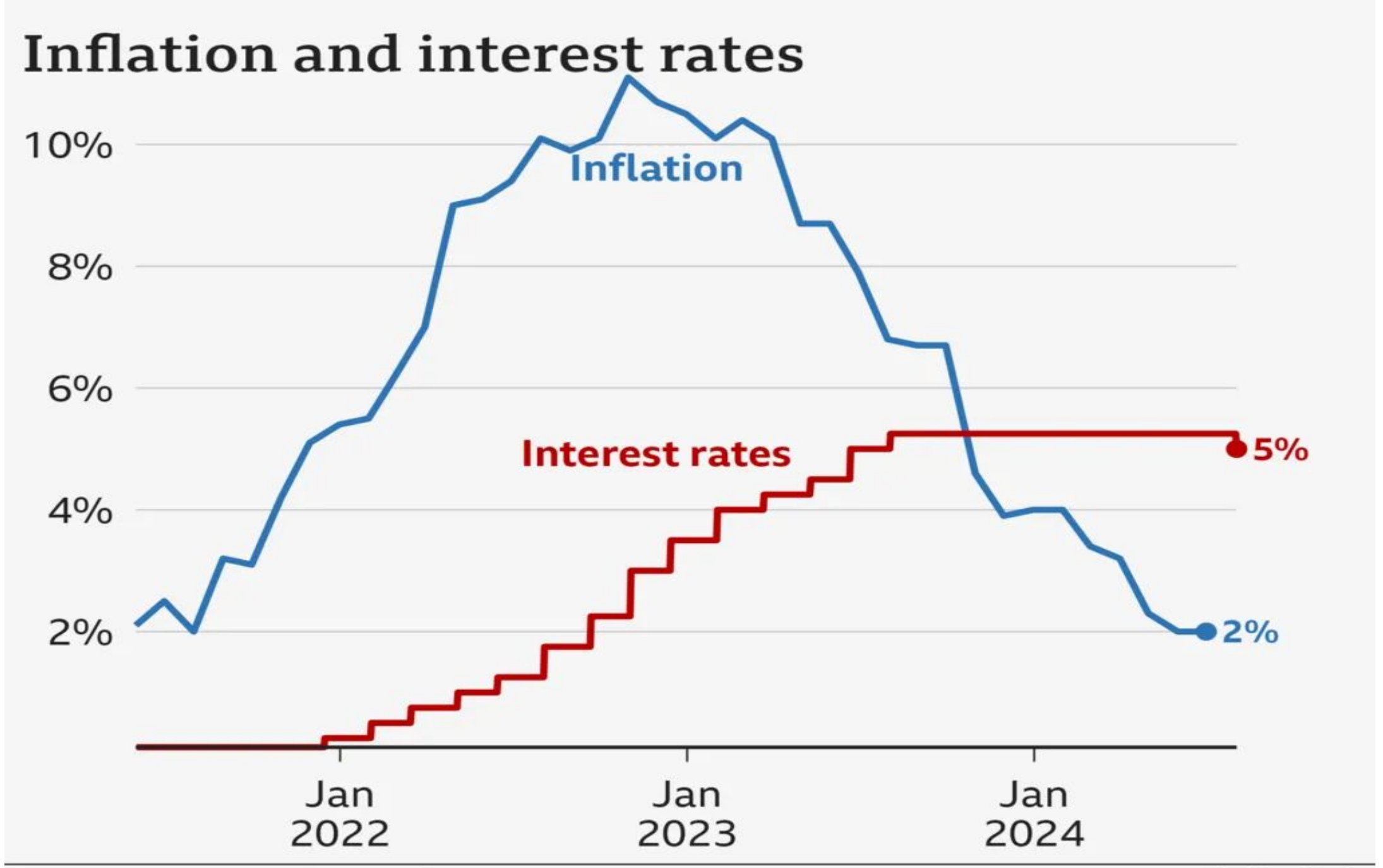
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What will a Federal Reserve interest rate cut mean for you?



Guardian graphic. Sources: Federal Reserve, BLS.

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Inflation and interest rates
10%
8%
6%
4%
2%
Inflation
Interest rates
5%
2%
Jan 2022
Jan 2023
Jan 2024
Source: ONS, Bank of England. Last update: 1 Aug 2024
BBC

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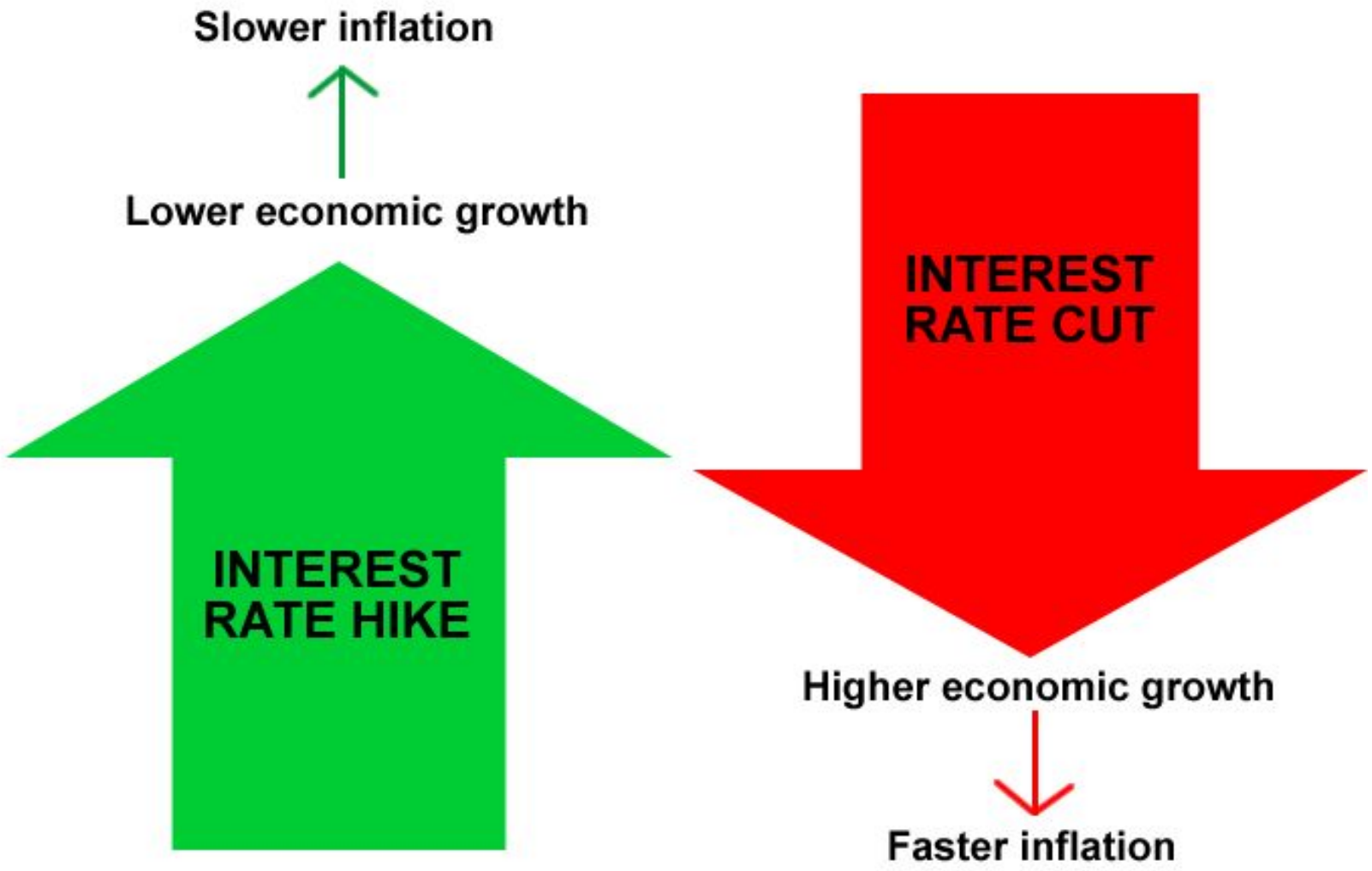
Ask Questions in speaker notes:

Podcast: Inflation Target Acquired

1. What is the inflation target set by the Federal Reserve?
2. How does the central bank influence inflation rates?

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Interest rates

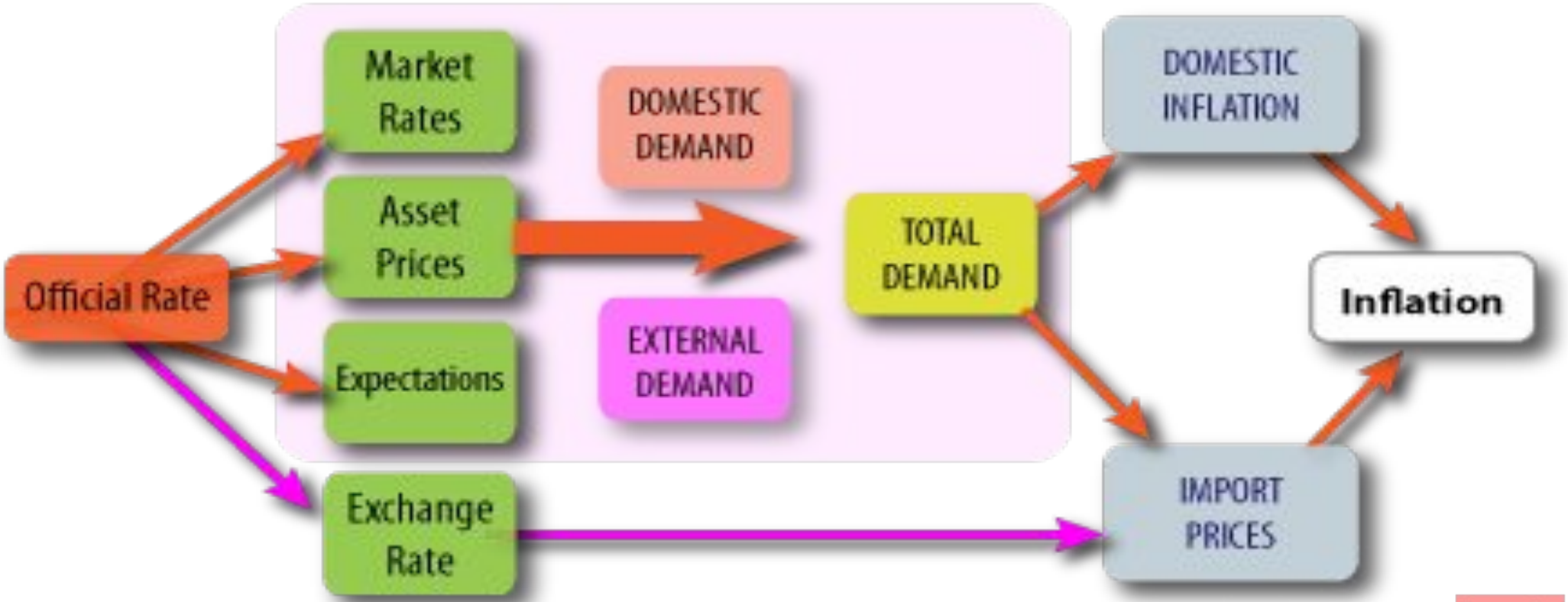


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Interest rates

Monetary Transmission Mechanism

This is the name for the **process** by which changes in the **Bank Rate** feeds through to changes in the **price level** and **inflation**.



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Monetary transmission mechanism

Effects on components of aggregate demand

$$AD = C + I + G + (X - M)$$

Consumption: Higher interest rates reduce consumption.

- **Mortgage** re-payments increase – less to spend on other goods.
- **Reduce asset prices** – negative wealth effect.
- **Saving** is more attractive.
- Buying goods on **credit** is more expensive (durable goods).

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Monetary transmission mechanism

Effects on components of aggregate demand

$$AD = C + I + G + (X - M)$$

Investment: Higher interest rates reduce investment.

- **More expensive to borrow for investment.**
- **More attractive to earn interest by saving in the bank.**
- **Opportunity cost is greater.**

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Monetary transmission mechanism

Effects on components of aggregate demand

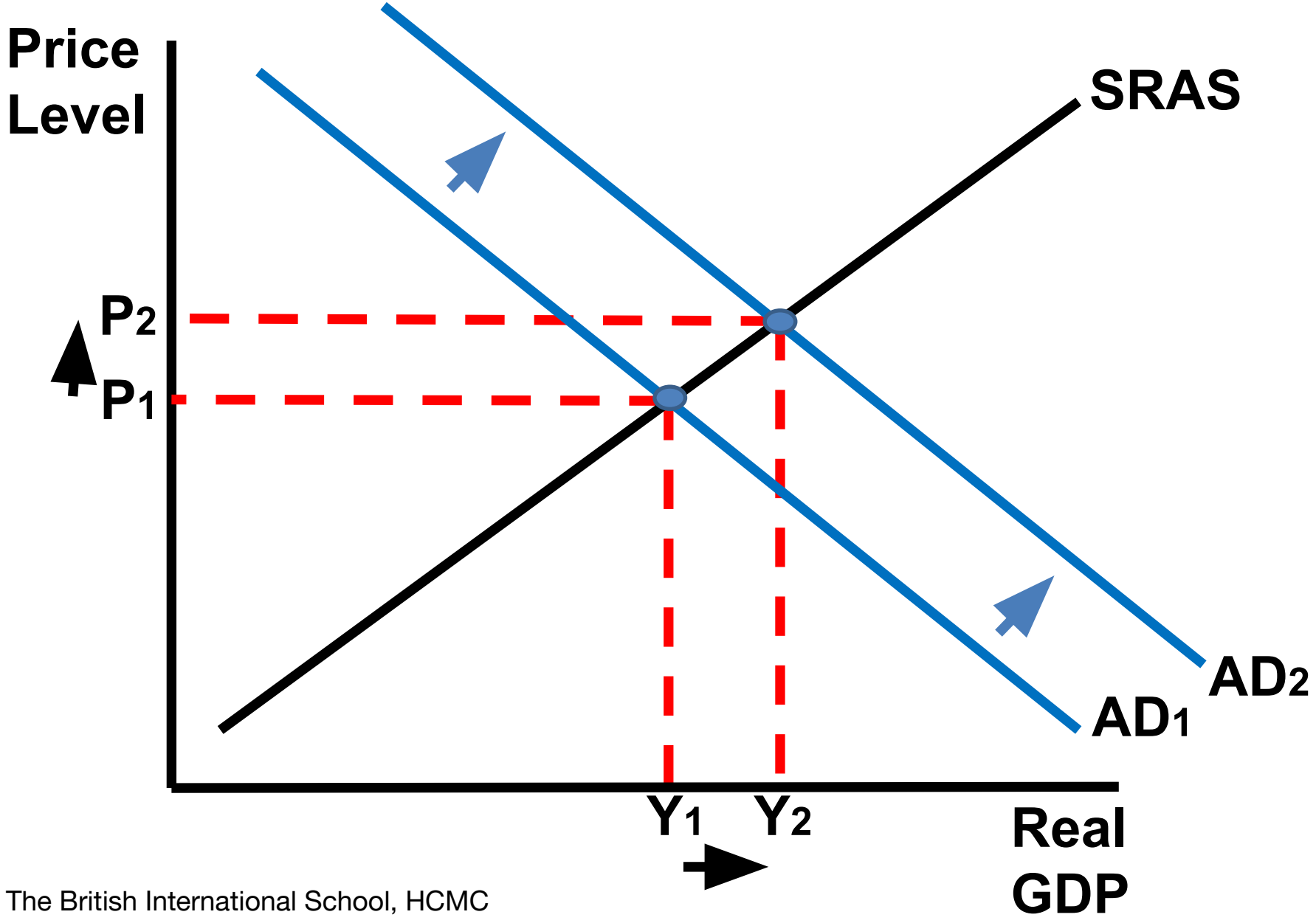
$$AD = C + I + G + (X - M)$$

Net Trade (X-M): Higher interest rates reduce net trade.

- **Encourages holding of pounds.**
- **Currency appreciates.**
- **Imports cheaper.**
- **Exports relatively more expensive for foreign buyers.**

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Monetary transmission mechanism



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Monetary transmission mechanism

Effects on components of aggregate supply

- Changes in interest rates may also have an impact on Long-run **Aggregate Supply**.

- But remember, this will **take time** and is **likely to be less** than the effects on the aggregate demand side.

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Evaluation

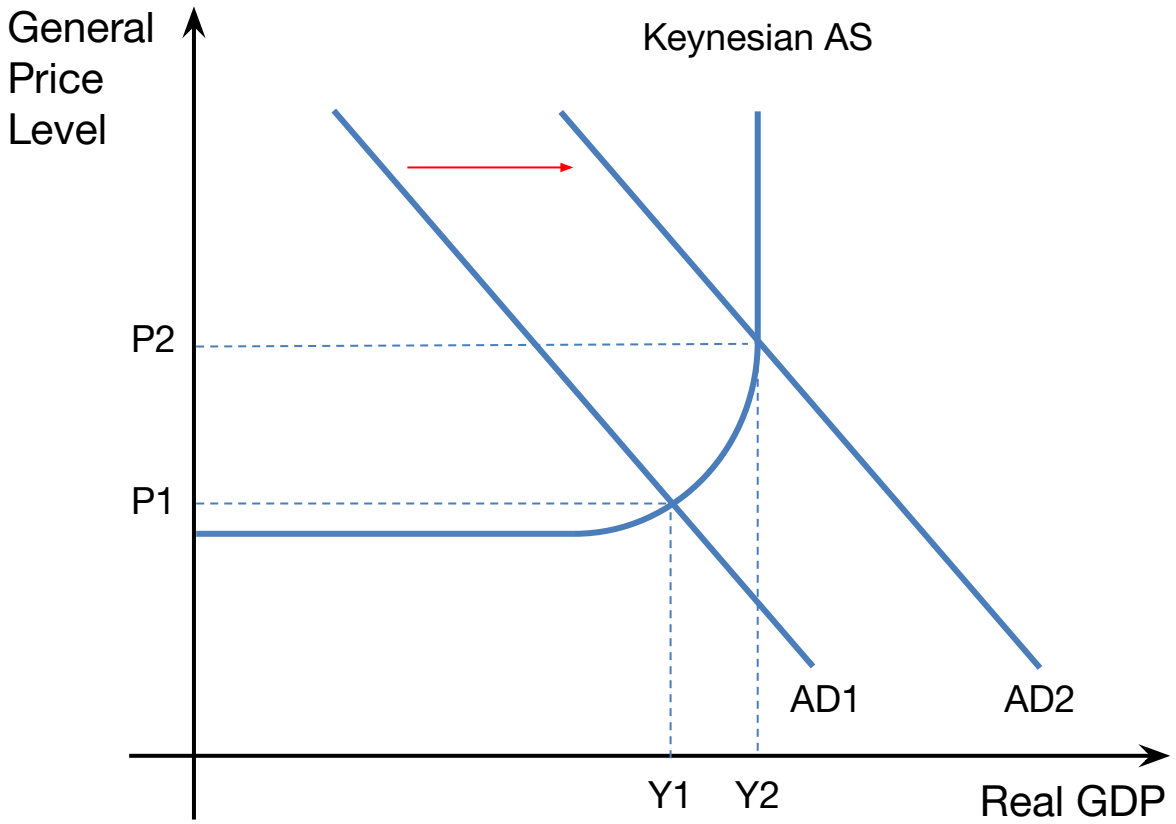
The **effectiveness** and **consequences** of monetary policy depends on a number of factors. When asked to evaluate in examination questions remember to evaluate in the **context of the question** being asked.

The following general points may be useful to consider:

- The **magnitude** of the changes in interest rates
- The **time lags** involved – changes in interest rates can take up to **18 months** to fully feed through to the **real economy** and the **price level**.
- The state of the economy and whether the economy is at or close to **full employment**, or whether there is a significant **output gap** and **spare capacity** in the economy. Is the economy at the **elastic** section or the **inelastic** section of the **Aggregate Supply curve**?

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use this diagram to show spare capacity influence



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Evaluation

- **Other changes** in the economy which may **offset** or **exacerbate** the expected changes arising from changes in interest rates.
- What is happening to **fiscal policy** and **supply-side policies**?
- Availability of **credit** – for households and for firms (credit crunch).
- The **magnitude** of the changes to the individual components of AD.

When evaluating, remember to **use the data** and **information** provided in the question to make **concrete evaluation points**.

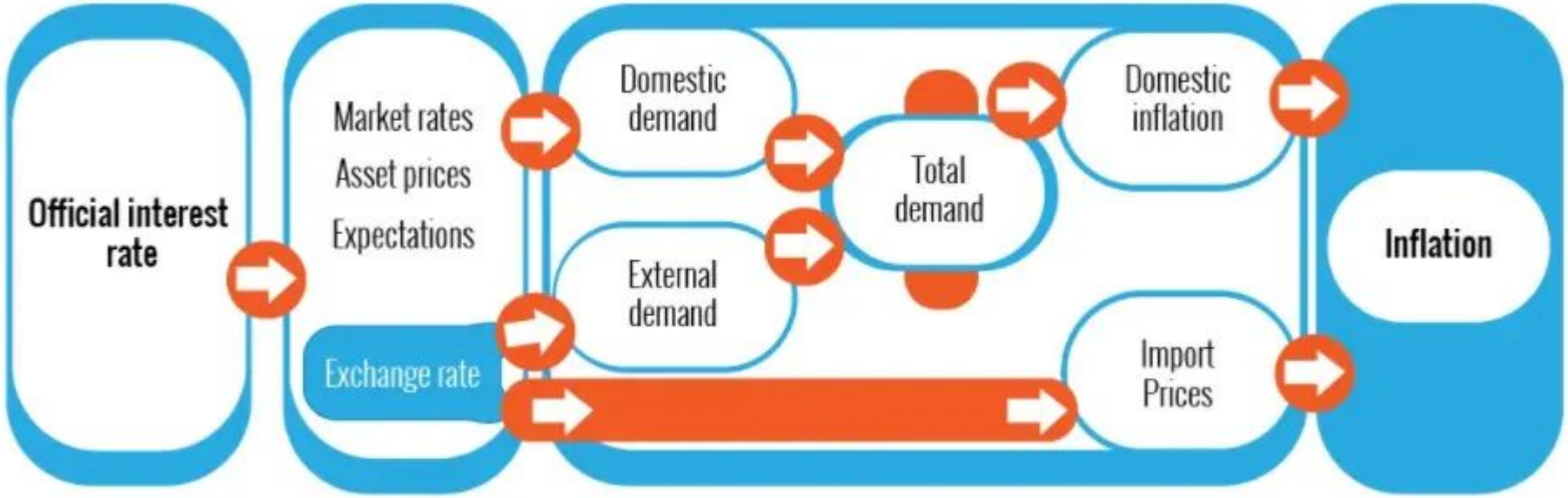
Avoid making **general comments** such as, “it depends on whether the changes in interest rates are large or small”.

Comment on the **actual figures** given and if they are significant or not.

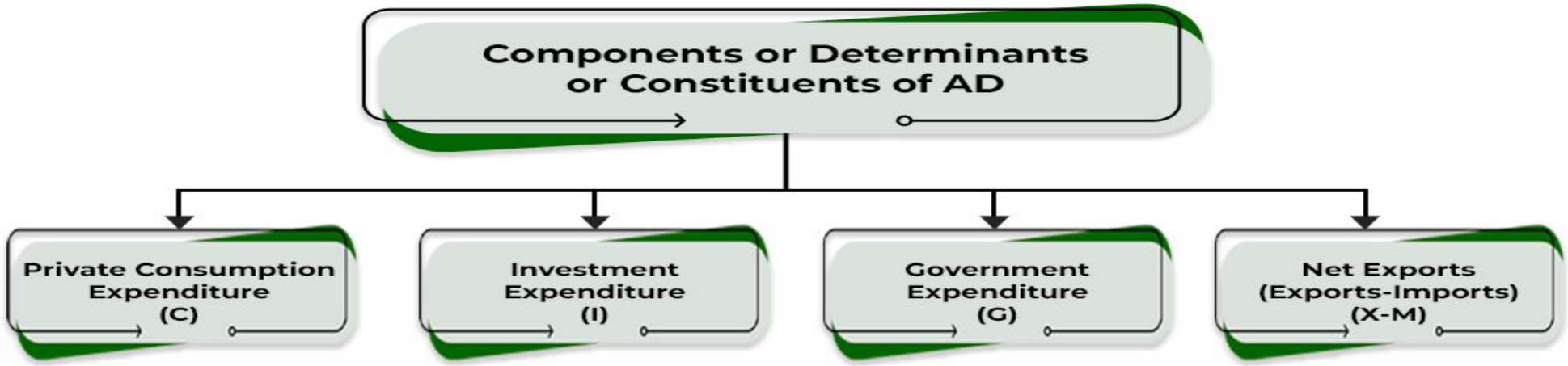
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Bank of England's transmission of monetary policy



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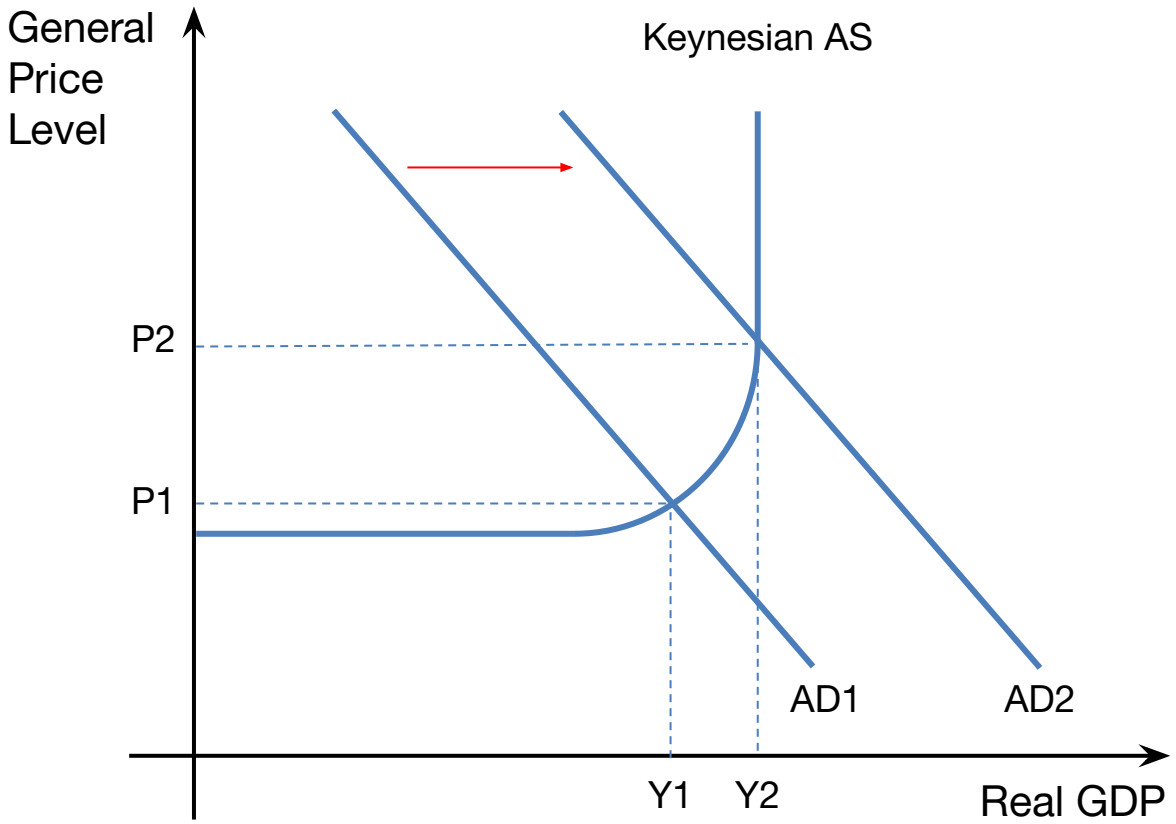


$$AD = C + I + G + (X-M)$$

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Goals of monetary policy



Low unemployment

The central bank may stimulate the economy by reducing interest rates. This reduces the cost of borrowing for firms and households, encouraging investment and consumption.

- AD increases from AD1 to AD2.
- Real GDP increases from Y1 to Y2.
- Unemployment decreases.

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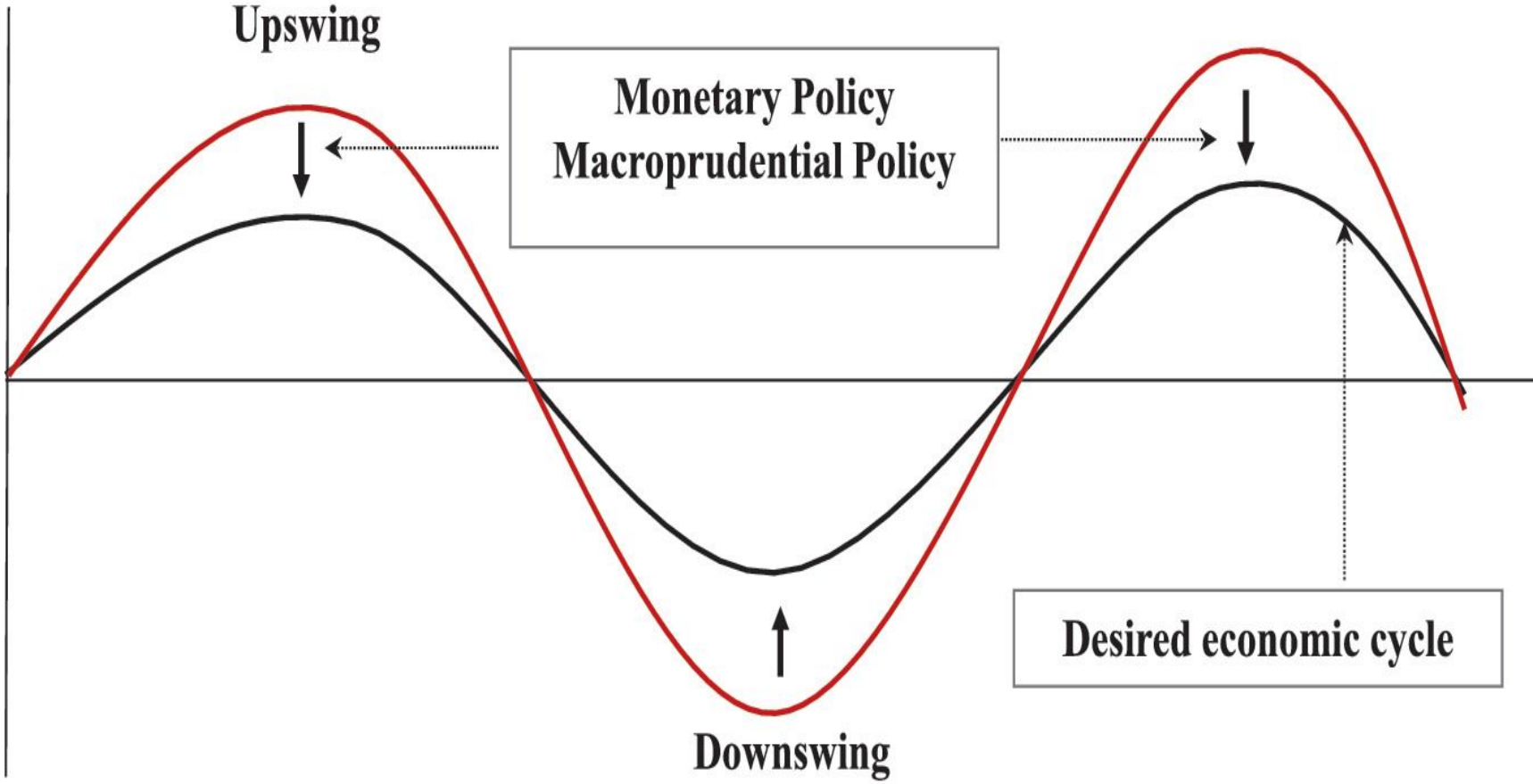
Goals of monetary policy

Reduce business cycle fluctuations

Monetary policy is used to **influence** the level of **economic activity**.

At times of an **economic downturn**, **interest rates** are **lowered** to **stimulate** the economy.

When the economy is **booming**, **interest rates** are **raised** to **lower inflationary pressure**.

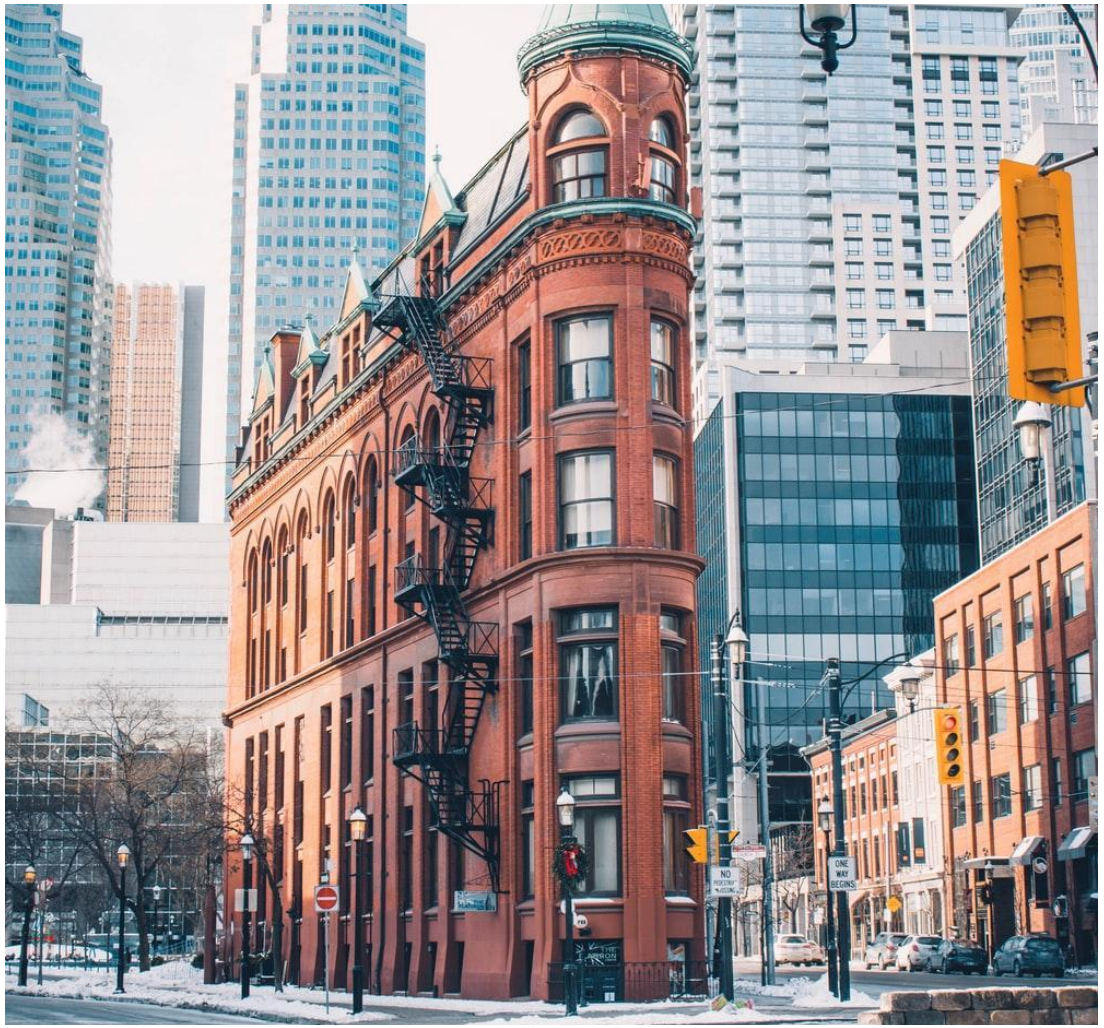


Using an AD/AS diagram, illustrate how the central bank influences the economy by raising interest rates.

However, there are issues with time lags:

- Recognition lags
- Response/Implementation lags
- Transmission lags

Goals of monetary policy



Promote a stable economic environment for a long-term growth

Low and stable inflation creates an environment of **economic stability.**

This generates a higher level of **certainty** and **confidence** for households and firms to **plan economic activities.**

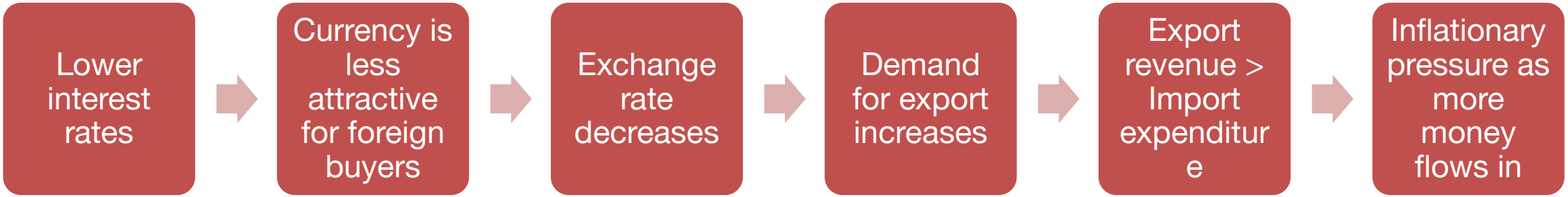
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Goals of monetary policy

External balance - (Balance of Payments – exports and imports)

The **external balance** refers to the value of an economy's **export revenue** being equal to its **import expenditure**.

Interest rates can influence the **exchange rate**, which in turn affects the value of exports and imports.



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The process of money creation by commercial banks (HL only)

Video: How Banks Create Money

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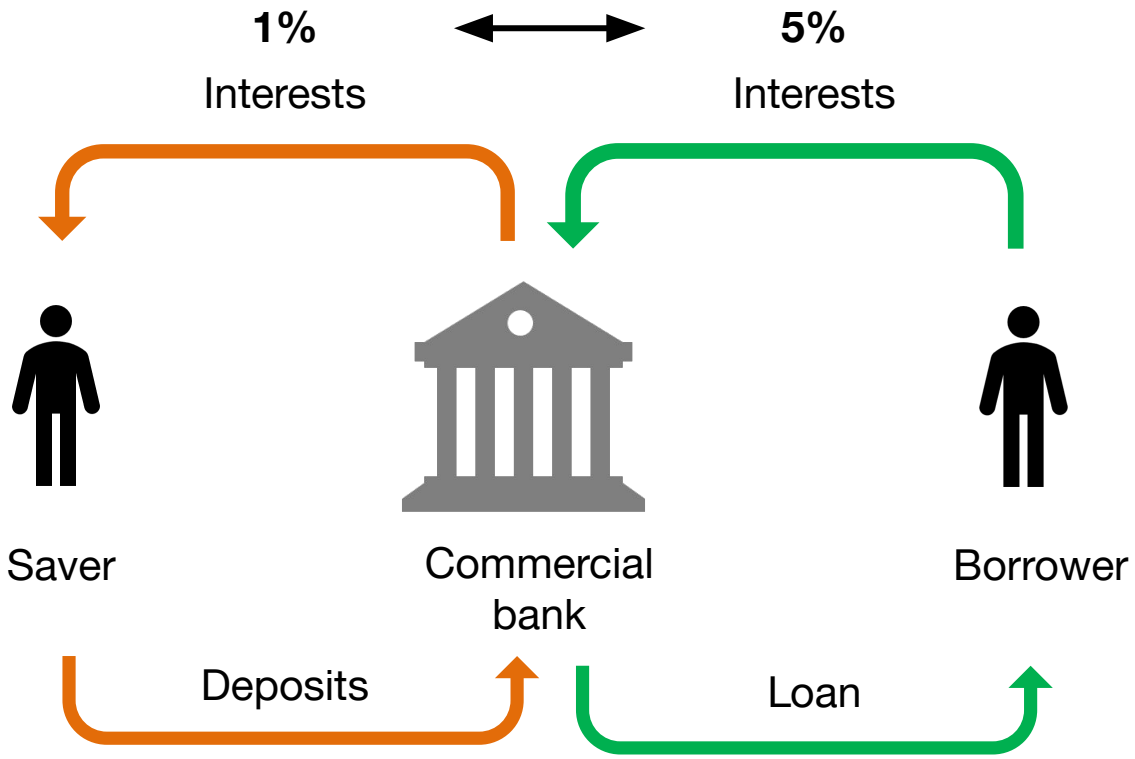
Credit creation (HL only)

Credit creation is the process by which banks create money by providing loans using saving deposits.

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The process of money creation by commercial banks (HL only)

- **Savers deposit money** in banks and gain **a rate of interest.**
- Banks will **create loans** from the deposits and lend some money to the **borrowers.**
- Banks **charge** an **interest rate** to **borrowers**.
- Interest rate charged to borrowers **exceed** the Interest rate paid to savers.
- This is how **banks profit** from providing loans.

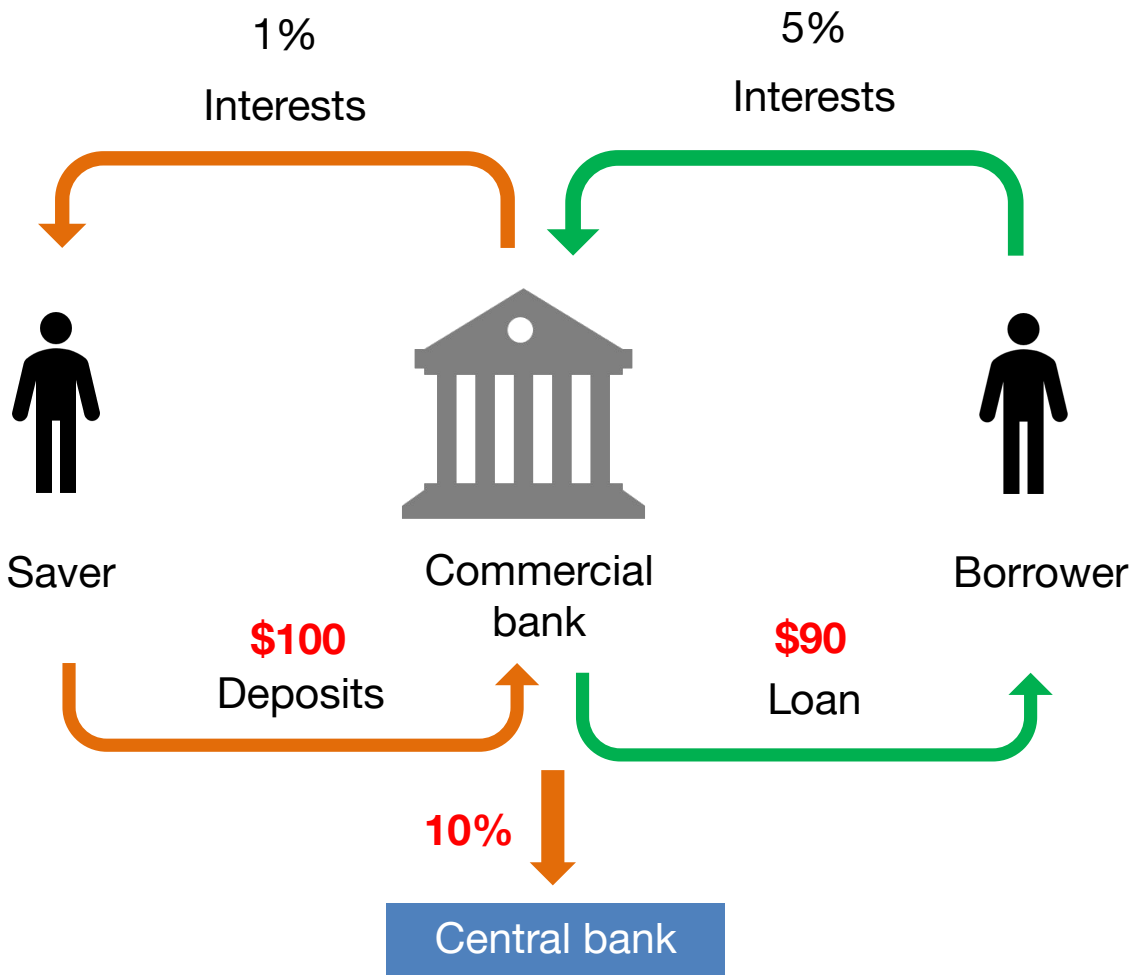


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The process of money creation by commercial banks (HL only)

For example,

- A saver deposits **$100** into the bank.
- The central bank decides a **minimum reserve ratio**, which is the required **percentage of deposits** commercial banks must keep in their vaults.
- Minimum reserve ratio, **for example 10%**
- The bank keeps **$10 of deposits** in their vaults and **lends** the remaining **$90 as loans.**



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The process of money creation by commercial banks (HL only)

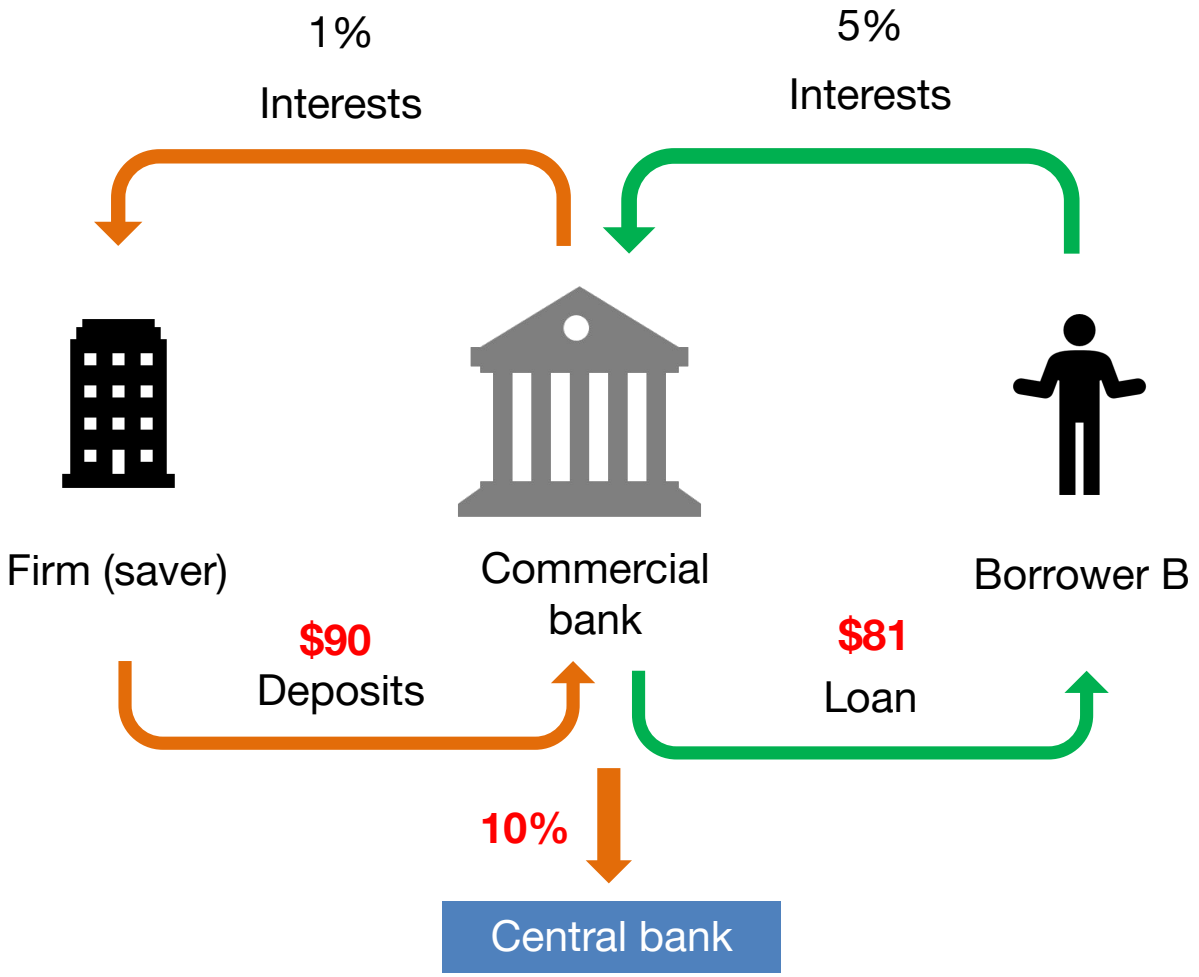
- The borrower uses the **$90 for consumption**, contributing to the economy and creating income for businesses.



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The process of money creation by commercial banks (HL only)

- The firm then **deposits the $90** earned into the **bank.**
- The **money supply increases** by **$90.**
- The **central bank reserves 10%** of the deposits.
- The **commercial bank lends** the **remaining $81** to another borrower.
- The process continues.
- **Deposits generate loans, which then returns to the bank as deposits, which subsequently increases money supply**.



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How Banks Create Credit

To illustrate how credit and bank deposits are created by the banking system we will make some simplifying assumptions:

- A **single monopoly bank**.
- Cash as the only **reserve asset** to meet customer withdrawals.
- A **10%** reserve asset (cash) ratio.

A customer now deposits **£1000 cash** with the bank.
For the bank, both its **assets and liabilities** increase by £1,000.

Assets	Liabilities
Cash: £1,000	Deposit: £1,000
Total Assets: £1,000	**Total Liabilities: £1,000**

Currently the bank has a **cash ratio of 100%.** It holds cash which covers **100% of its liabilities**.

How Banks Create Credit

Assets	Liabilities
Cash: £1,000	Deposit: £1,000
Total Assets: £1,000	**Total Liabilities: £1,000**

The bank can now use this cash to **issue credit** (loans) to other customers. With a **10% cash ratio**, the bank can lend up to **£9,000** to other customers.

Assets	Liabilities
Cash: £1,000 Advances: £9,000	Deposit: £1,000 Created deposits: £9,000
Total Assets: £10,000	**Total Liabilities: £10,000**

The process of money creation by commercial banks (HL only)

Money multiplier (supplementary)

A formula used to calculate how much an initial deposit increases the economy's money supply by:

$$\text{Money multiplier} = \frac{1}{\text{Reserve ratio}}$$

In the previous example, the money multiplier = $\frac{1}{0.1}$ = 10.

We can then calculate how much the initial deposit ($100) increases the money supply:

(savings x money multiplier) – savings = impact on money supply

($100 x 10) – $100 = $900

The initial $100 of savings increases the money supply by $900.

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The process of money creation by commercial banks (HL only)

Money multiplier (supplementary)

Suppose the central bank wants to limit the economic activity and reduce inflationary pressures. The reserve ratio increases to 20%. Calculate how much an initial saving of $100 would increase money supply by.

Money multiplier = $\frac{1}{0.2}$ = 5

Money supply increased = ($100 × 5) – $100 = $400



How is the reserve ratio and money supply related?

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Demand and supply of money (HL only)



The **demand for money** (D_m) refers to the willingness and ability of borrowers to obtain loans.

Stakeholders who may demand loans includes:

- Households
- Firms
- Government

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Demand and supply of money (HL only)



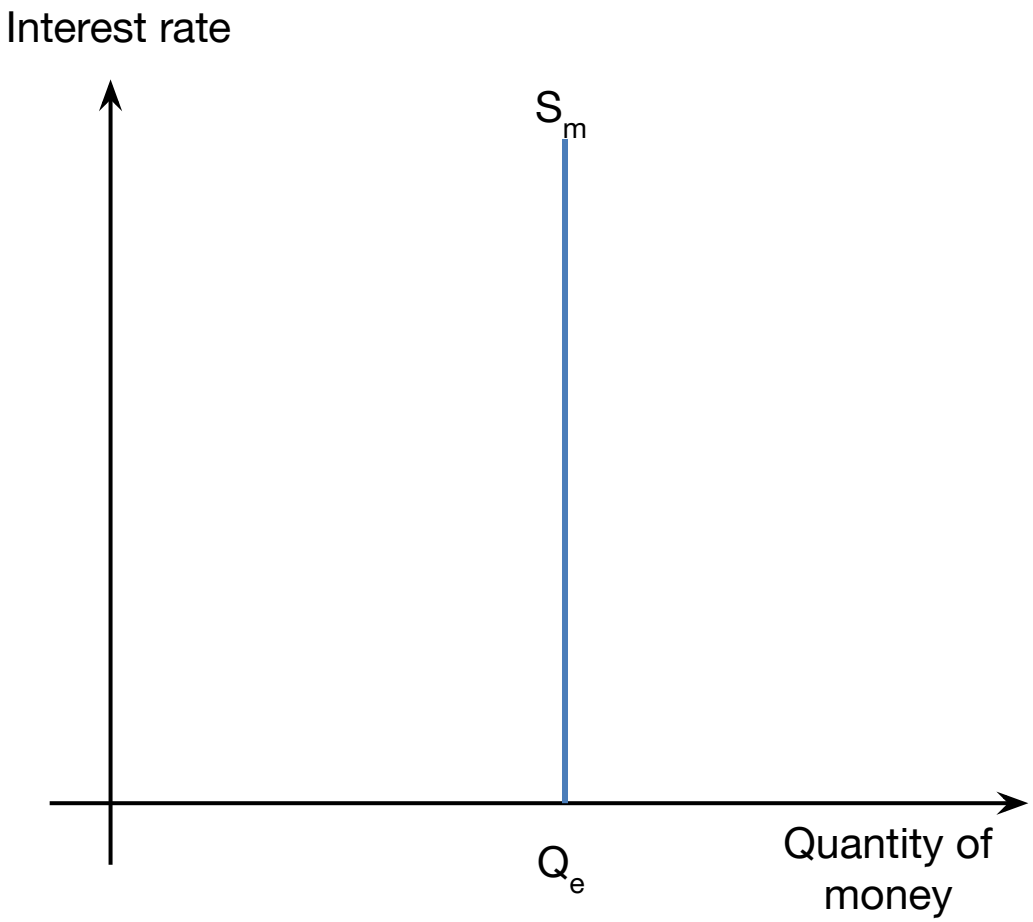
The **demand for money** (D_m) shows the inverse relationship between interest rates and the quantity of money/loans demanded.

As interest rates increase, the quantity of loans demanded decrease, because it is more expensive to borrow money. The opposite applies when interest rates decrease.

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Demand and supply of money (HL only)



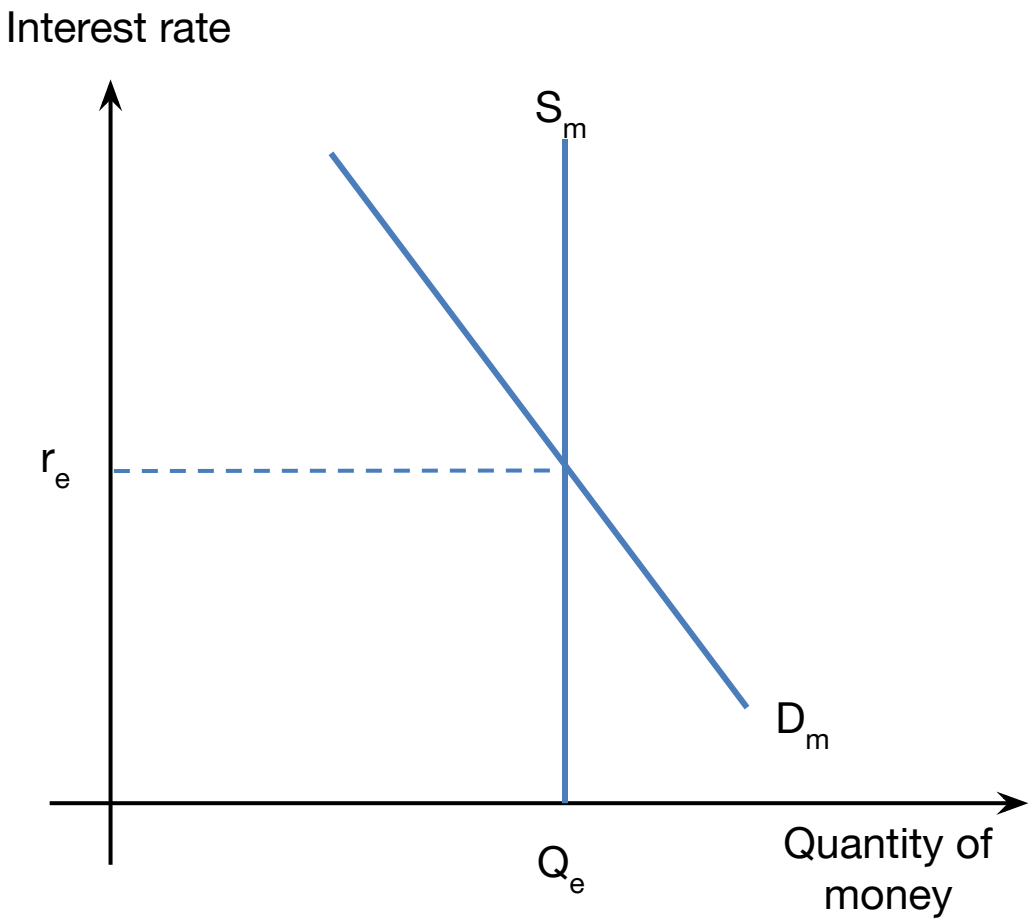
The **supply of money** (S_m) refers to the amount of money circulating in the economy at any time.

Why is the supply curve of money perfectly inelastic?

The supply of money is fixed by the central bank (monetary authority) at any point in time and is not dependent on interest rates.

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Demand and supply of money (HL only)



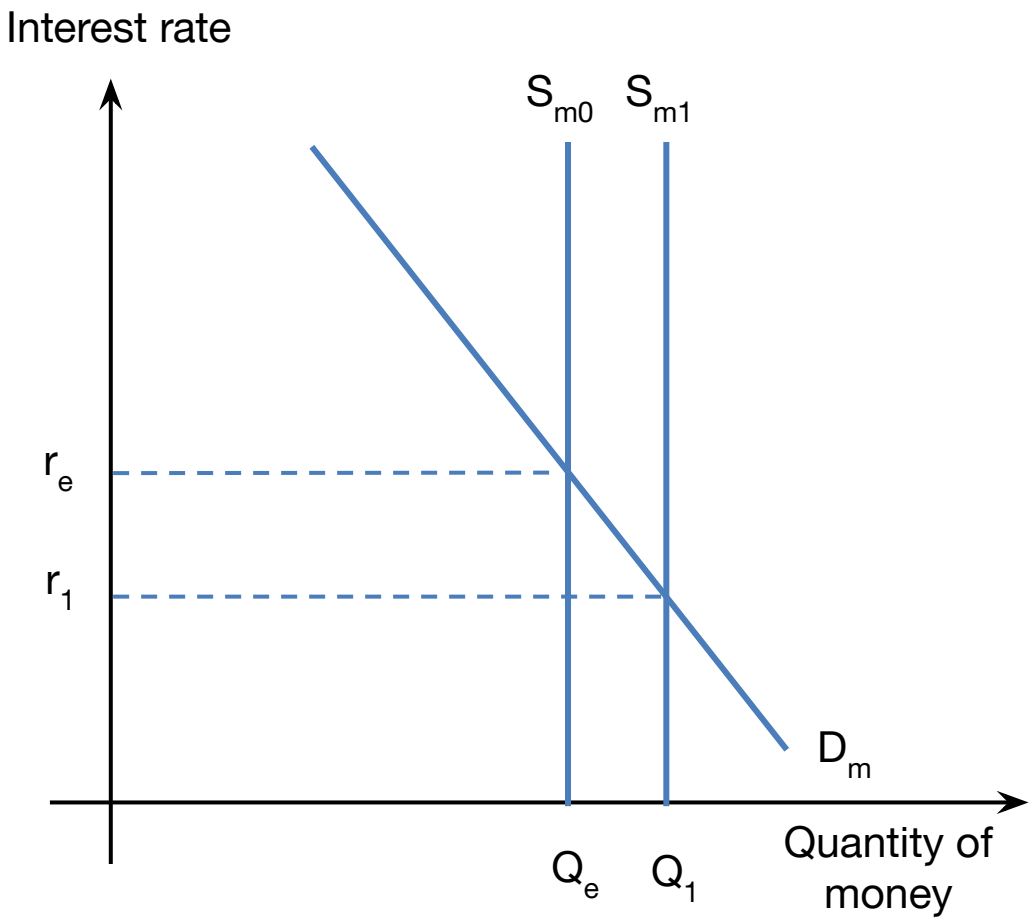
The **equilibrium interest rate** (r_e) can be determined by the intersection of the demand (D_m) and supply (S_m) of money.

This is the market interest rate at which borrowers and lenders agree on.

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Demand and supply of money (HL only)

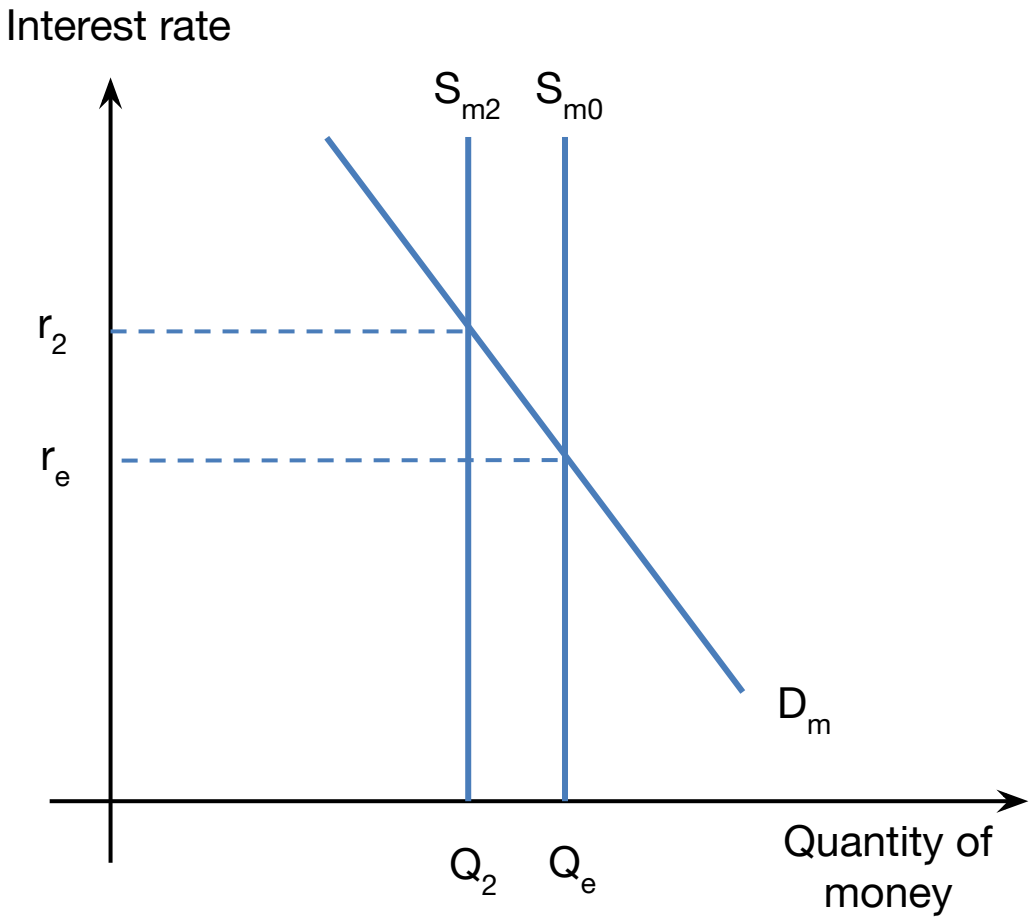


Increases in money supply

The central bank may increase money supply using the tools of monetary policy (outlined in slide 32) from S_{m0} to $S_{m1.}$ This decreases the interest rate to r_1.

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Demand and supply of money (HL only)



Decreases in money supply

The central bank may decrease money supply using the tools of monetary policy (outlined in slide 32) from S_{m0} to $S_{m2.}$ This decreases the interest rate to r_2.

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Demand and supply of money (HL only)

The central bank considers the following factors when setting interest rates:

- **The exchange rate –** Lower interest rates decreases the demand of the domestic currency. The exchange rate will fall, encouraging the sale of exports.
- **Property prices –** Property prices directly impact the level of consumer confidence and the potential economic growth of the economy.
- **The rate of growth and nominal wages –** Higher wage rates imply that firms will increase prices. A rise in interest rate helps to reduce inflationary pressure.
- **The state of the economy –** A reduction in interest rates lifts the economy out of a recession.
- **Business confidence levels –** Lower interest rates encourage firms to invest.

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Tools of monetary policy (HL only)

There are four main methods used by the central bank to influence the money supply and interest rates. These include:

- Open market operations
- Minimum reserve requirements
- Changes in the central bank minimum lending rate
- Quantitative easing (QE).

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Open market operations (HL only)

Open market operations refers to the buying and selling of government bonds by a central bank to control money supply and interest rates.

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Open market operations (HL only)

Governments may sell bonds to investors. In exchange for the investor's money, a bond certificate offers interest payments and the promise to repay the initial investment at a certain date. This is used to raise finance for operations and public projects. There are two ways in how open market operations occurs:

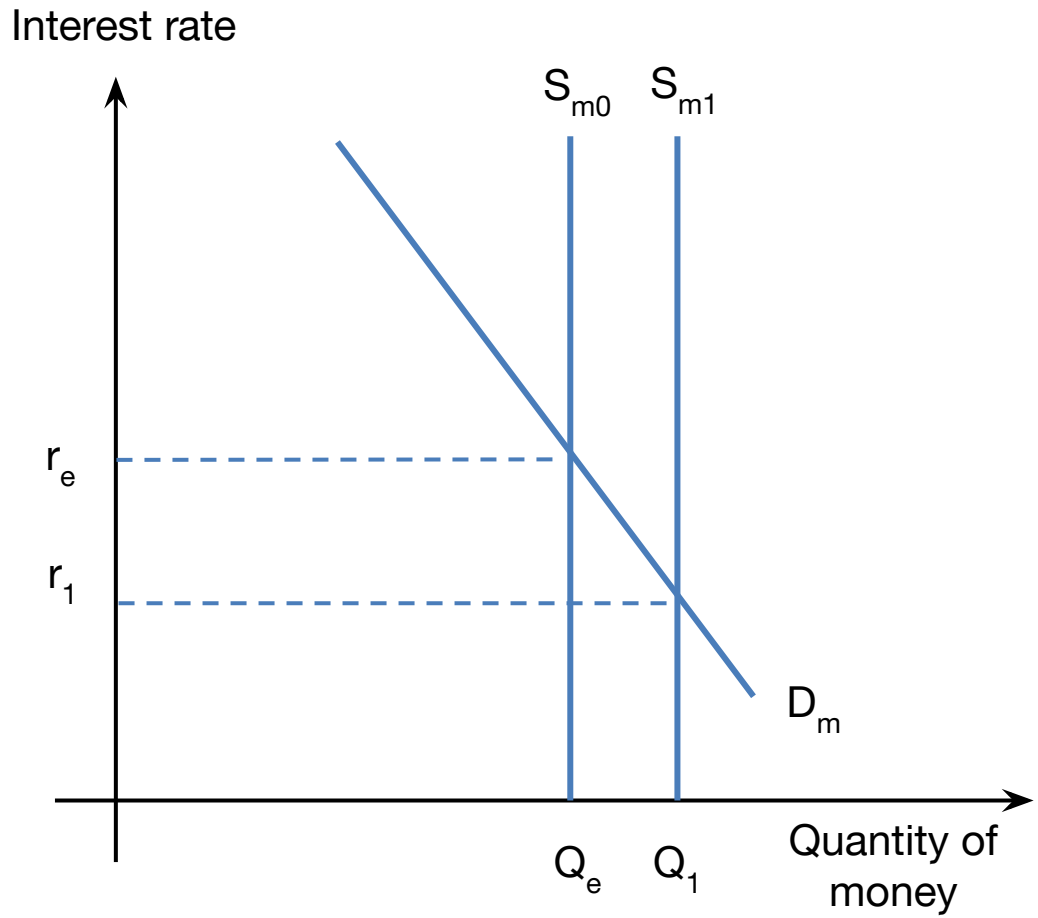
- Central bank buys government bonds to increase money supply.
- Central bank sells government bonds to decrease money supply.

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Open market operations (HL only)

If a central bank wishes to carry out **expansionary monetary policy** and aims to lower interest rates, they could buy government bonds from commercial banks.

This increases money supply from Sm0 to Sm1 as commercial banks have now have additional money to create more loans. Interest rates decrease from re to r1.



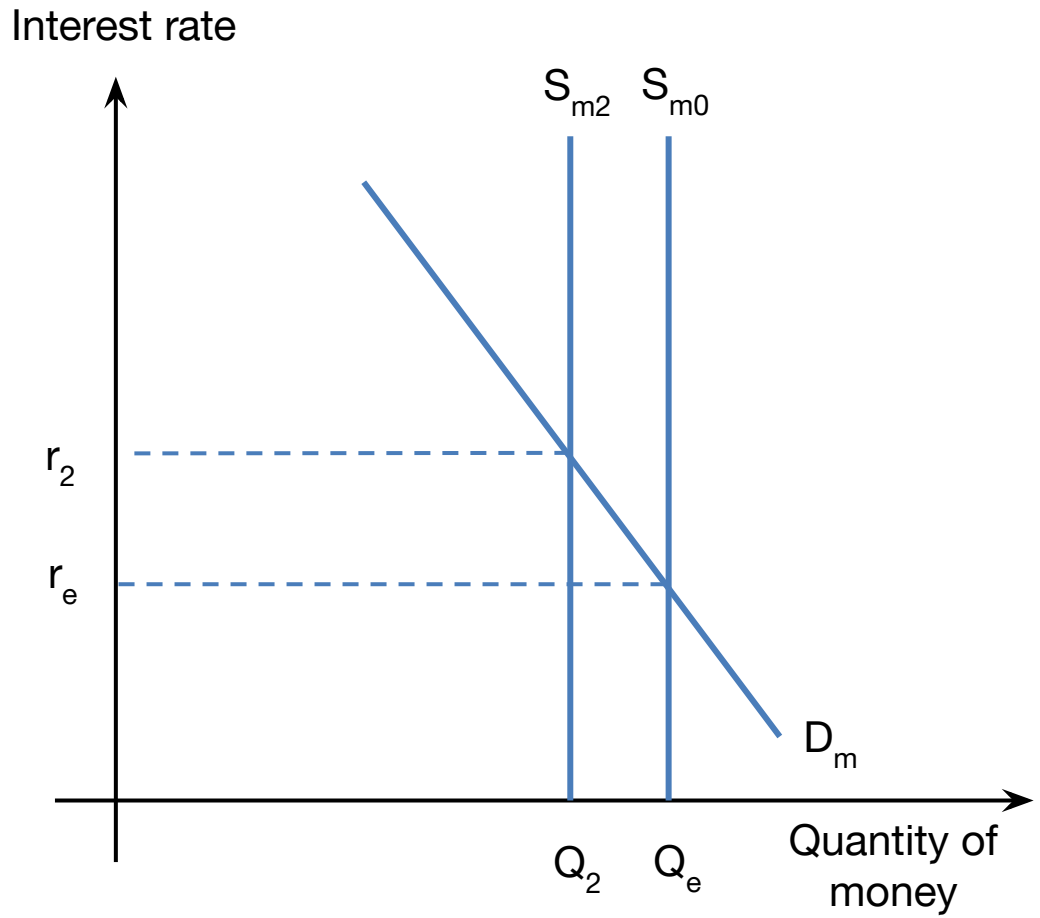
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Open market operations (HL only)

If a central bank wishes to carry out **contractionary monetary policy** and aims to increase interest rates, they could sell government bonds from commercial banks.

This decreases money supply from Sm0 to Sm2 as commercial banks have now have less money loan out. Interest rates increase from re to r2.



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Minimum reserve requirements (HL only)

The **minimum reserve requirement (MRR)** is the required percentage of deposits that commercial banks keep in their vaults.

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Minimum reserve requirements (HL only)

There is an inverse relationship between MRR and the money multiplier. The higher the MRR, the lower the money multiplier, meaning that there is a __lower__ amount of money created.



What is the relationship between money supply and interest rates?

Expansionary monetary policy:

Lower MRR to increase money supply, injecting liquidity to the economy. Interest rate decreases, raising consumption and investment expenditure. This stimulates economic growth.

Contractionary monetary policy:

Raise MRR to limit the money supply, thereby increasing interest rates. Consumption and investment expenditure decrease, lowering inflationary pressure.

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Minimum lending rate (HL only)

The minimum lending rate is the interest rate charged by the central bank on loans to commercial banks.

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Changes in the central bank minimum lending rate (HL only)

The MLR influences all interest rates on bank loans, credit transactions, and mortgages.

Expansionary monetary policy:

The central bank can lower the MLR. The commercial banks will also lower their interest rates paid to savers, encouraging consumption and investment through borrowing. Hence, AD increases.

Contractionary monetary policy:

The central bank can raise the MLR. The commercial banks will also increase their lending rates, which encourages people to save more. Consumption and investment hence decrease, lowering the level of AD.

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Case study (HL only)



Article: Fed keeps rates near zero

1. How does the change in MLR affect the other loan rates and stimulate the economy?
2. How does the change in MLR benefit borrowers?
3. Which stakeholders are worse off as a result of low interest rates?

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Quantitative easing (HL only)

In times of **recessions**, even though **interest rates** are **near zero**, banks may be **reluctant to lend** while consumers are **unwilling to borrow and consume** due to **low confidence** in the economy.

This leads to the **liquidity trap**, when the **MLR** cannot be further lowered, and **expansionary monetary policy is ineffective.**

The central bank may then use another method to **stimulate** the economy: **quantitative easing**.

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Quantitative easing (HL only)

Quantitative easing is a tool of monetary policy in which the central bank **injects money directly** into the economy through **purchasing government or corporate bonds.**

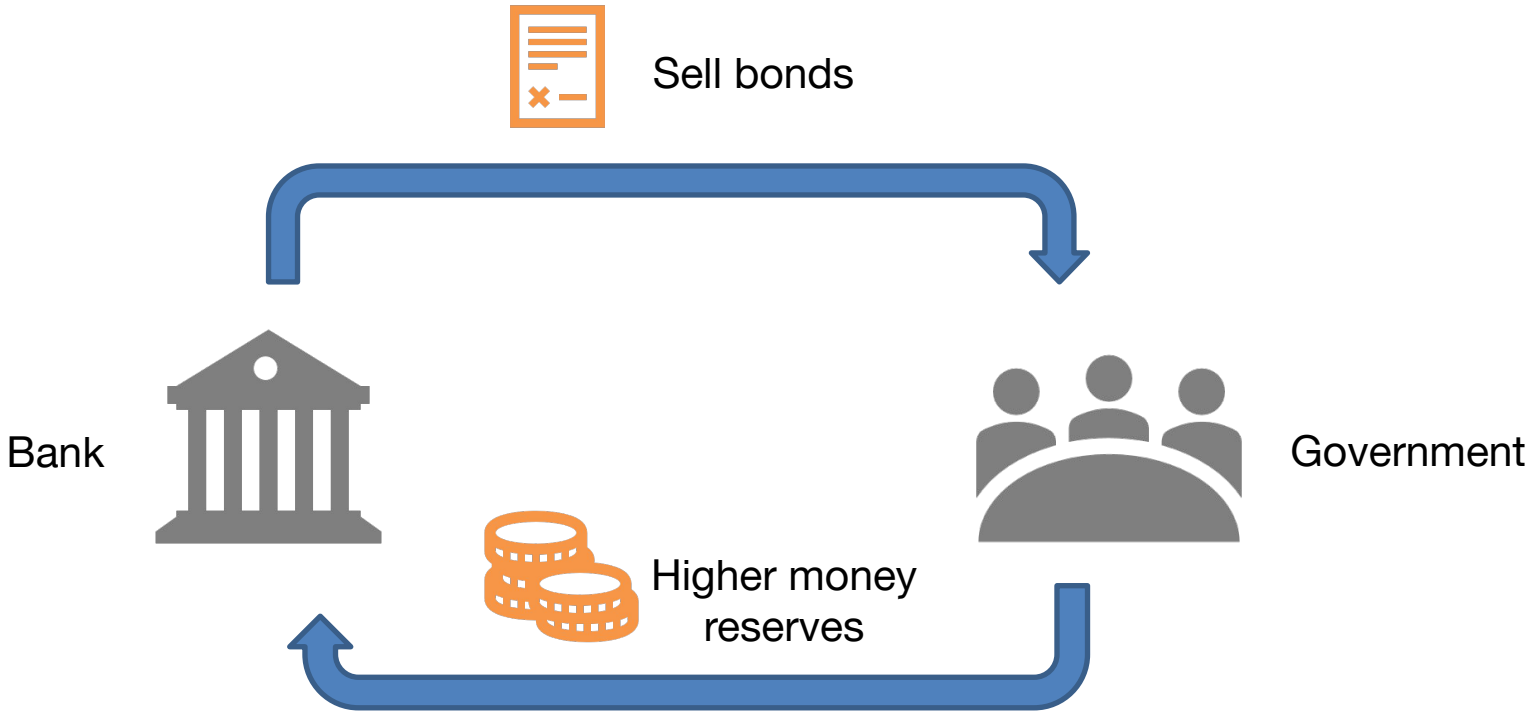
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Quantitative easing (HL only)

Bonds are a type of debt. Banks selling corporate bonds to the government will receive additional money which **increases liquidity**.

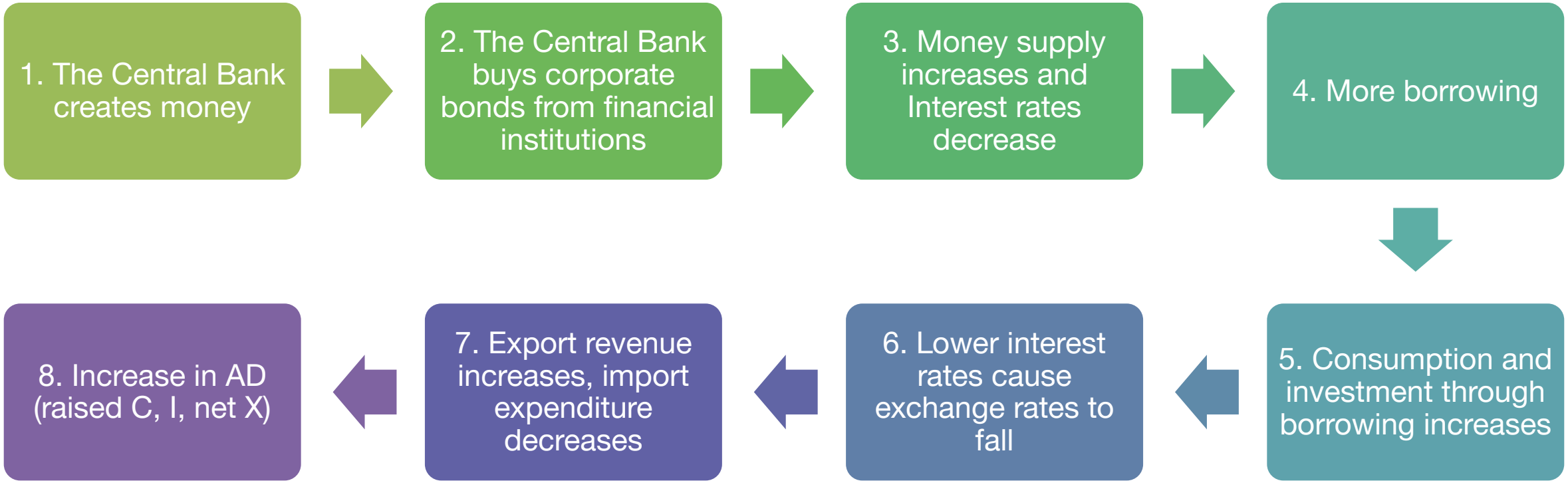
In other words, **money supply increases**, encouraging lending and hence economic growth.

QE also helps to **avoid deflation.**



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Quantitative easing (HL only)



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Summary of the tools of monetary policy (HL only)

The table shows the actions of the central bank in expansionary and contractionary monetary policies.

	Open market operations	Minimum reserve requirements	Minimum lending rates	Quantitative easing
Expansionary	Purchase government bonds	Lower MRR	Lower MRL	Purchase corporate bonds
Contractionary	Sell government bonds	Raise MRR	Raise MRL	-

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Real versus nominal interest rates

Interest rate is the cost of borrowing and reward for saving of money, expressed as a percentage.

Nominal interest rate is the actual interest rate that is agreed between the bank and the customer, which is the rate that borrowers pay their loans or savers receive from their deposits.

Real interest rate considers the impact of inflation on the cost of borrowers and the return for savers.

Real interest rate = Nominal interest rate – Inflation rate

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Real versus nominal interest rates

Example and Interpretation

Justin saves $5,000 for a year and receives a nominal interest rate of 2%. The annual inflation rate is 1.5%. Calculate the nominal return and real return for Justin's savings after 1 year.

Nominal return = $5,000 × 2% = $100

Real interest rate = 2% – 1.5% = 0.5%

Real return = $5,000 × 0.5% = $25

Although Justin's savings increases by $100, the **real** return on his savings considering inflation is only $25.

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Real world example – data analysis

Source: Visualizing the 700-Year Fall of Interest Rates

Data Analysis Questions

1. What do you notice from the data?
2. What questions do you wonder about the data?
3. Research information that may help you answer your questions from Q2.
4. What conclusions can you make from Q1, Q2, and Q3?

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Real versus nominal interest rates

Negative interest rates can occur when the nominal interest rate is less than the inflation rate as

Real interest rate = Nominal interest rate – Inflation rate

For example, if the nominal interest rate is 1% and the inflation rate is 2%, then the real interest rate is -1%. This means the purchasing power of any savings would **fall** by 1% by year end.

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Real world example (supplementary)

Video: Negative Interest Rates: Explained

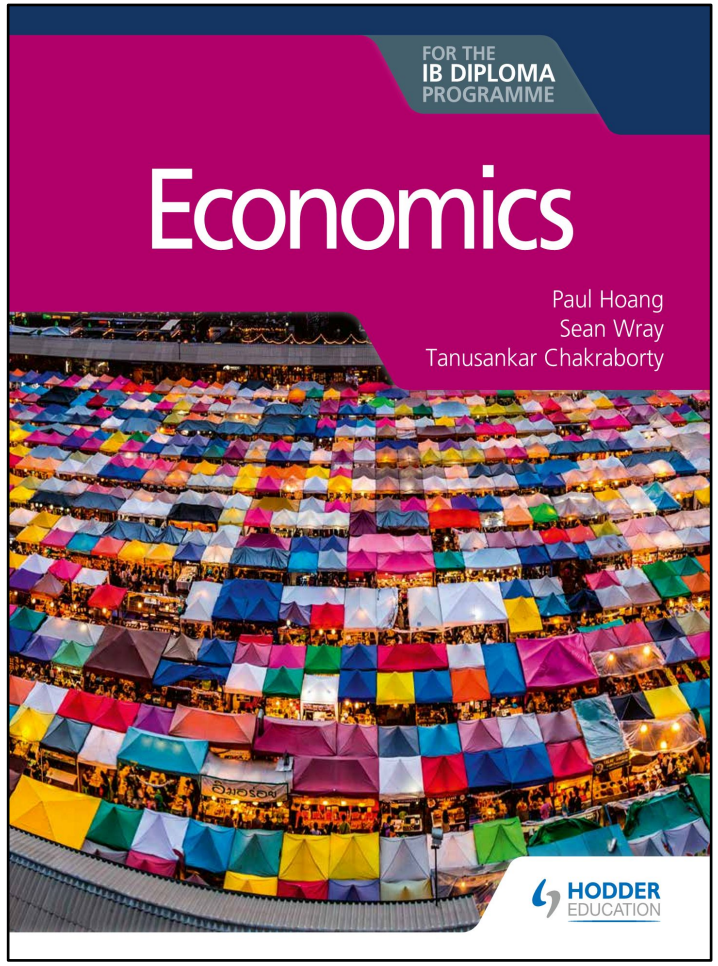
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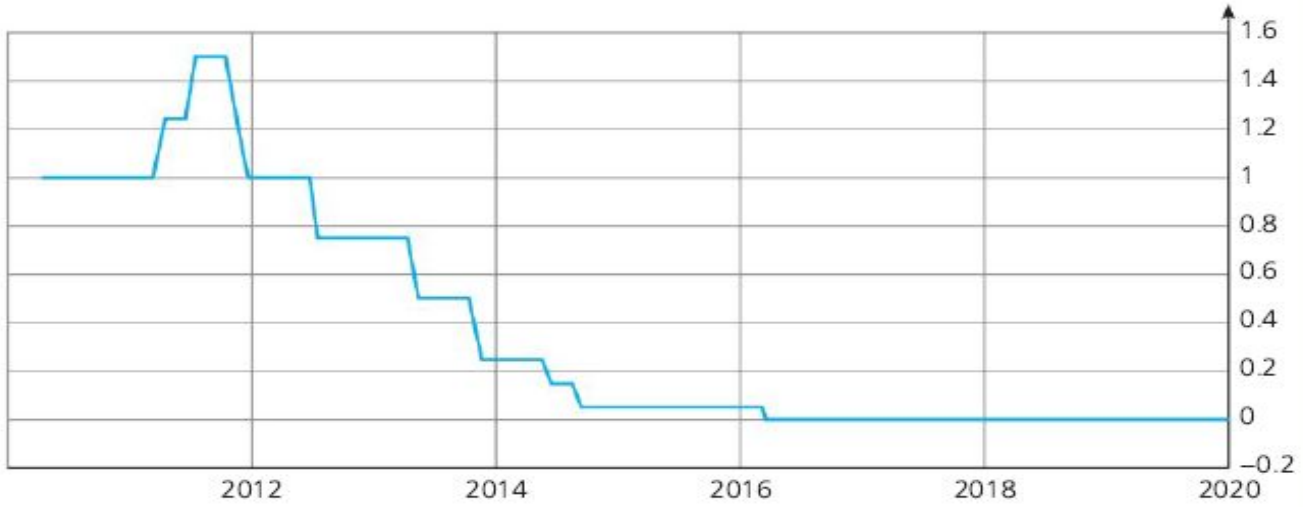
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- [4 marks]



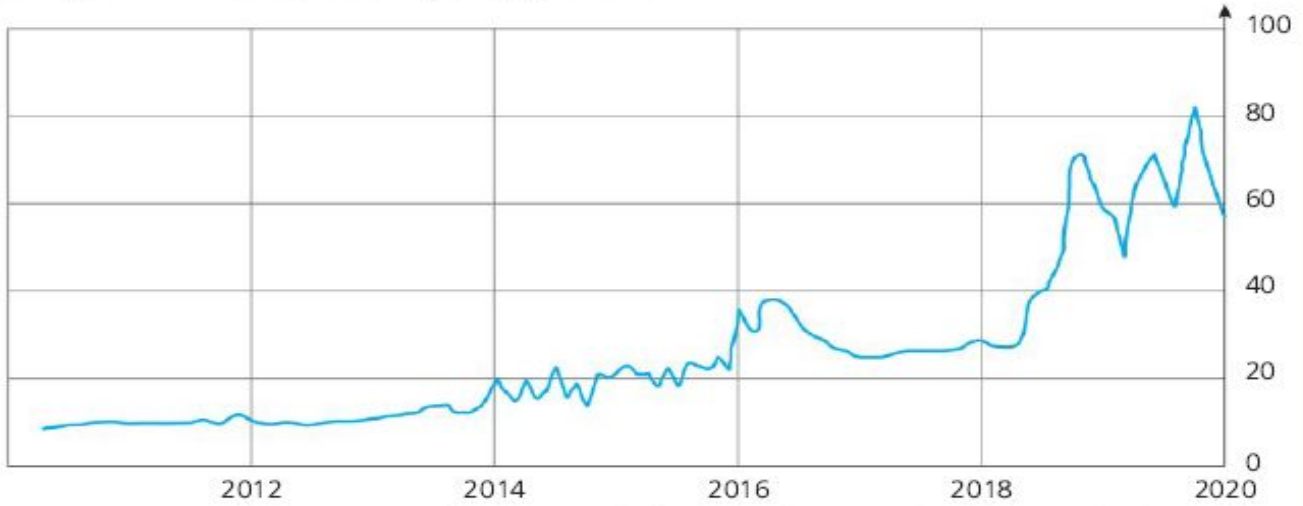
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PAPER 3 EXAM PRACTICE QUESTION 24.2 (HL ONLY)



Source: Trading Economics https://tradingeconomics.com/spain/interest-rate

Figure 24.11 Interest rates in Spain (%), 2010–20



Source: Trading Economics https://tradingeconomics.com/argentina/interest-rate

Figure 24.12 Interest rates in Argentina (%), 2010–20

With reference to each of the graphs above for Spain (Figure 24.11) and Argentina (Figure 24.12), explain why the respective governments have adopted contrasting interest rate policies. *[4 marks]*

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[Paper 3 Exam Practice Question 24.2 – HL only]

The contrasting use of interest rate policies in Spain and Argentina is likely to be linked to the respective economic conditions in these countries. For example, Spain's continual decline in interest rates to a ZIRP (zero interest rate policy) suggests the economy has been contracting – in fact, Spain struggled for a decade to recover from the global financial crisis of 2008.

By contrast, the Argentinian economy is likely to have suffered from high rates of inflation, so the use of higher interest rates (of over 80%) has been used to tackle such problems – in fact, Argentina has experienced double-digit inflation rates for many consecutive years.

Award *1–2 marks* for an answer that shows limited understanding of the use of interest rate policies to tackle economic issues. There is limited, if any, reference to the charts.

Award *3–4 marks* for an answer that shows good understanding of the use of interest rate policies to tackle economic issues, such as recession or inflation. There is effective use of the given charts.

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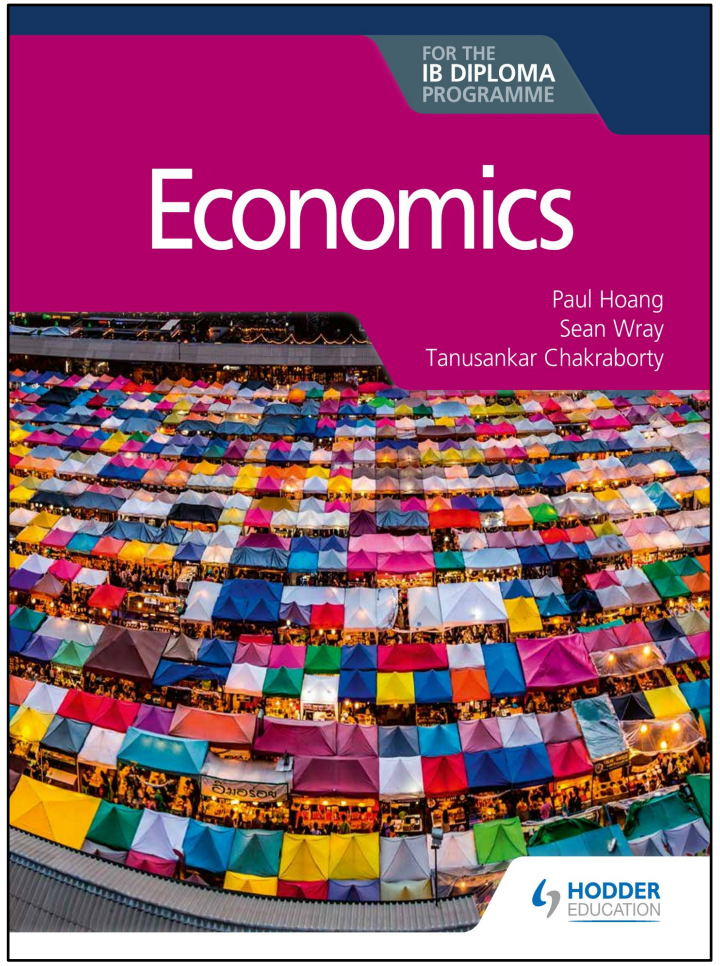
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- [3 marks]



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PAPER 2 AND 3 EXAM PRACTICE QUESTION 24.1

Use the data below to calculate the real interest rate in 2018, 2019 and 2020. *[3 marks]*

Year	Consumer price index (CPI)	Nominal interest rate (%)
2017	100	5.0
2018	110	5.5
2019	120	5.0
2020	115	4.5

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[Paper 2 and Paper 3 Exam Practice Question 24.1]

Year	Consumer price index (CPI)	Inflation rate (%)	Nominal interest rate (%)	Real interest rate (%)
2017	100		5.0	
2018	110	(110 – 100)/100 = 10.00	5.5	**–4.50**
2019	120	(120 – 110)/110 = 9.09	5.0	**- 4.09**
2020	115	(115 – 120)/120 = –4.17	4.5	**8.67**

Award *1 mark* for each correct answer, up to the maximum of *3 marks*. Apply the own figure rule (error carried forward) where appropriate.

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Expansionary monetary policy

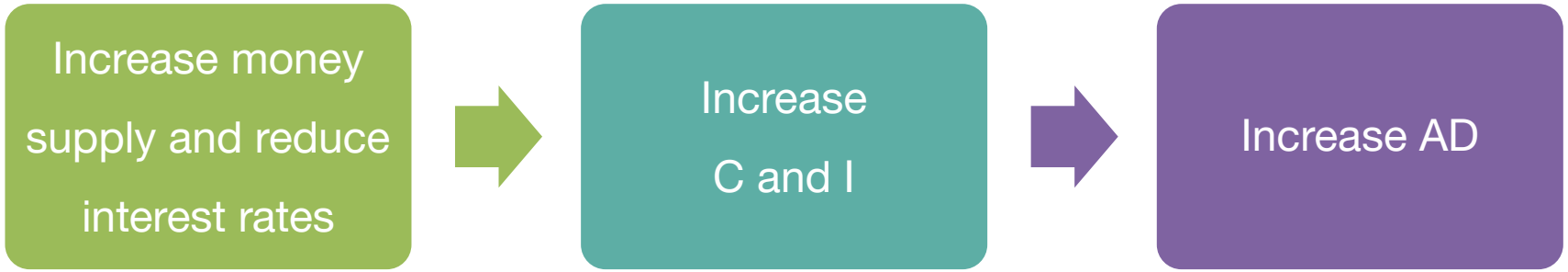
Expansionary monetary policy aims to increase the level of aggregate demand by increasing money supply and reducing interest rates.

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Expansionary and contractionary monetary policies

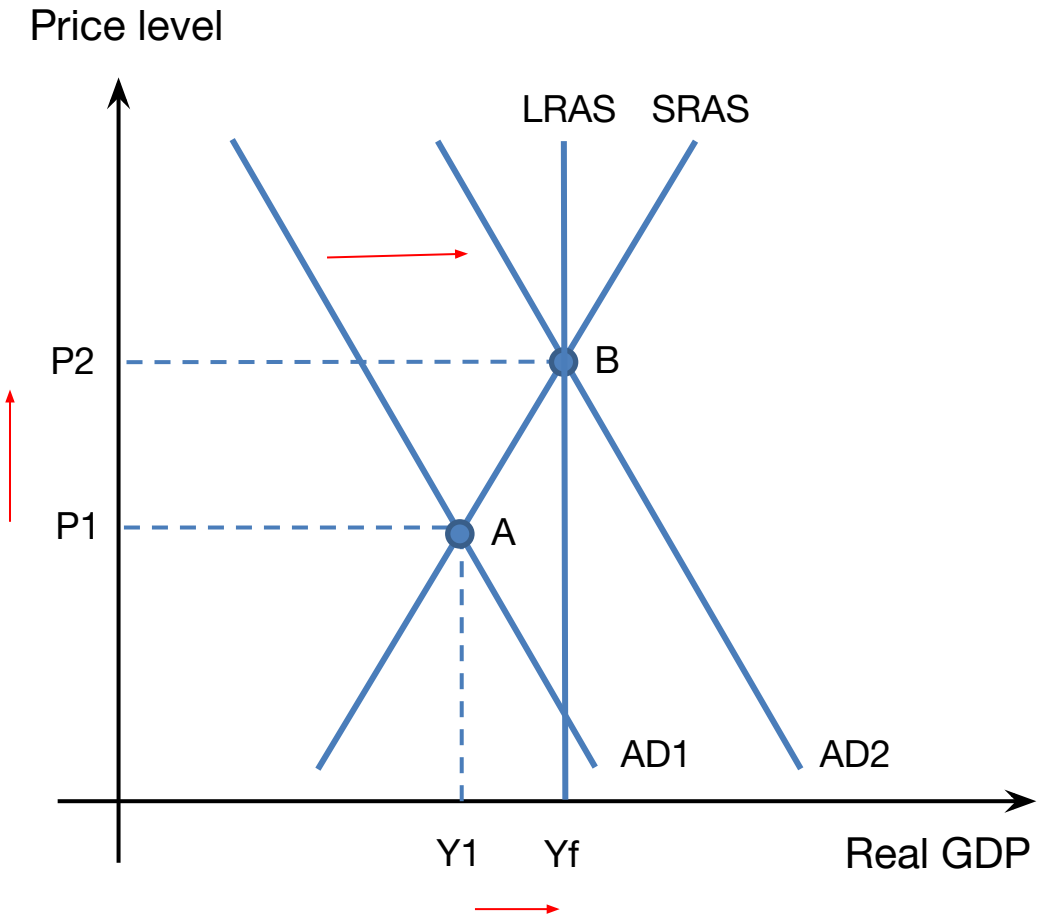
Expansionary monetary policy

The cost of borrowing and reward for saving reduces, increasing consumption and investment through increased borrowing and reduced savings. An expansionary monetary policy hence stimulates the economy by shifting the AD curve to the right.



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Expansionary and contractionary monetary policies



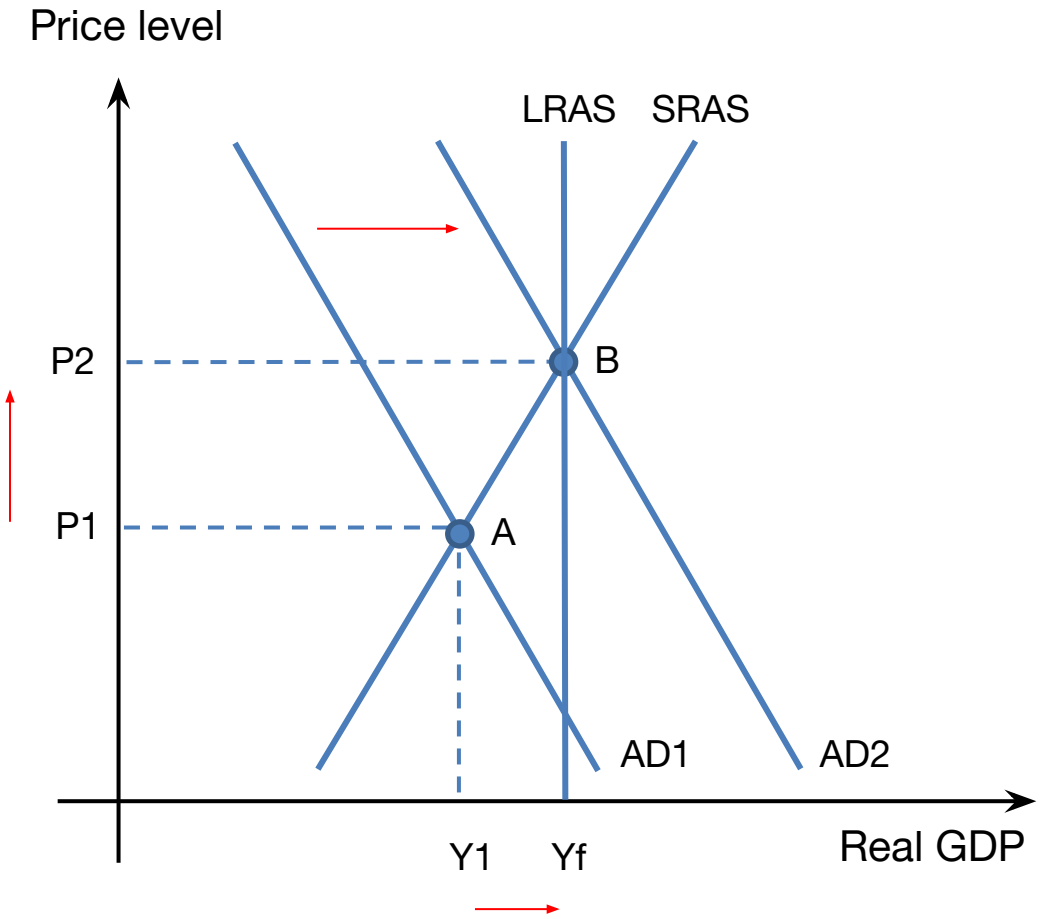
Expansionary monetary policy

- Consumption and investment increases
- AD curve shifts right from AD1 to AD2
- Equilibrium moves from point A to B
- Real GDP increases from Y1 to Yf, closing the deflationary gap
- Price level rises, leading to inflationary pressures.
- Unemployment reduces.

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Expansionary and contractionary monetary policies



When is expansionary monetary policy used?

- When inflation is below the target rate
- When the economy is suffering from a recessionary gap where unemployment is high and economic growth is low or negative.

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Contractionary monetary policy

Contractionary monetary policy aims to decrease the level of aggregate demand by decreasing money supply and increasing interest rates.

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Expansionary and contractionary monetary policies

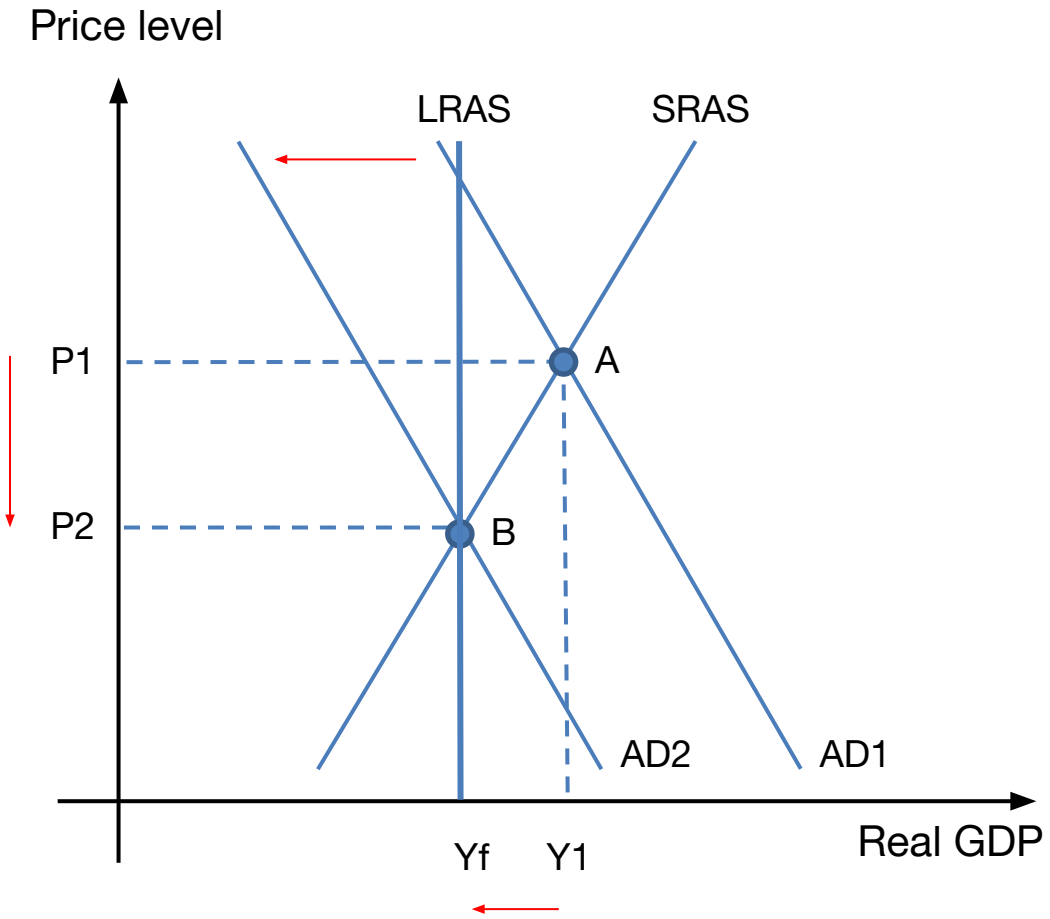
Contractionary monetary policy

The cost of borrowing and reward for saving increases, decreasing consumption and investment through decreased borrowing and increased savings. Contractionary monetary policy reduces or dampens AD and economy activity and subsequently reduces inflationary pressures.



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Expansionary and contractionary monetary policies

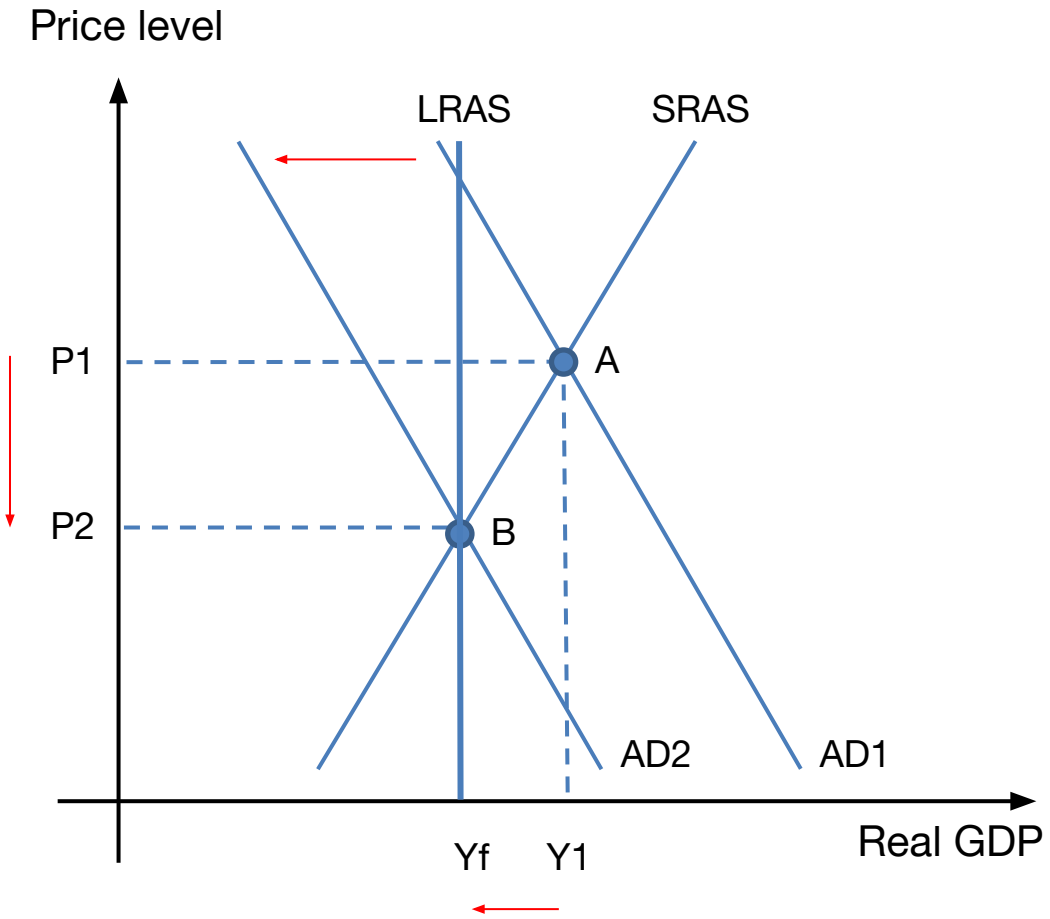


Contractionary monetary policy

- Consumption and investment decreases
- AD curve shifts left from AD1 to AD2
- Equilibrium moves from point A to B
- Real GDP decreases from Y1 to Yf, closing the inflationary gap
- Price level decreases from P1 to P2
- Unemployment increases

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Expansionary and contractionary monetary policies



When is contractionary monetary policy used?

- When inflation is above the target rate
- When the economy is experiencing high levels of economic growth and low levels of unemployment, contractionary monetary policy may be used as a proactive measure to slow AD growth and prevent the economy from overheating.

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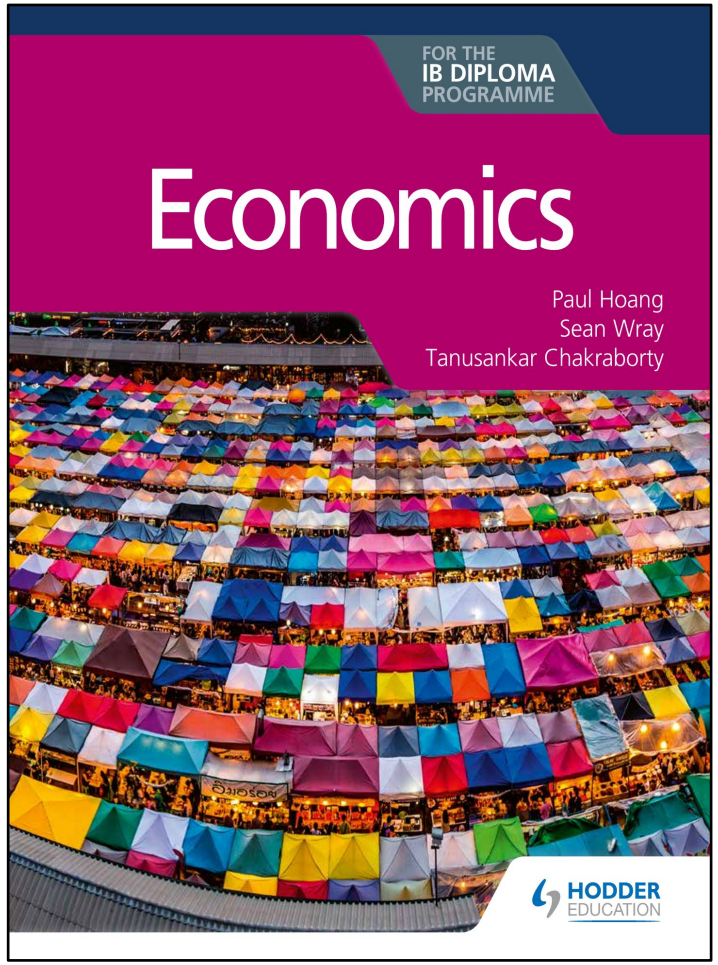
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- [4 marks]
- Paper 2 and 3 Exam Practice Question 24.4
- [4 marks]



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PAPER 2 AND 3 EXAM PRACTICE QUESTION 24.3

With the aid of an AD–AS diagram, explain how the use of expansionary monetary policy can help to close a deflationary (recessionary) gap. *[4 marks]*

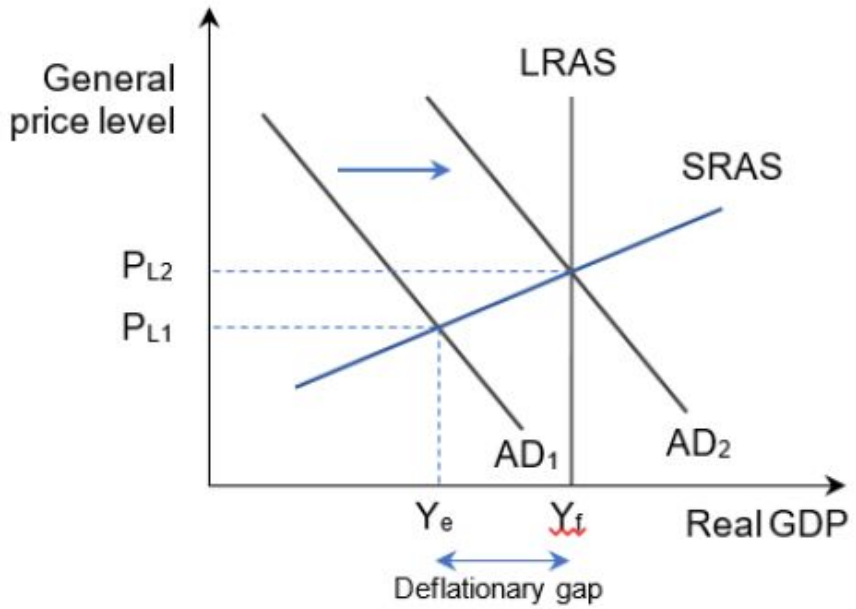
PAPER 2 AND 3 EXAM PRACTICE QUESTION 24.4

With the aid of an AD–AS diagram, explain how the use of contractionary monetary policy can help to close an inflationary gap. *[4 marks]*

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[Paper 2 and 3 Exam Practice Question 24.3]



An expansionary monetary policy (by reducing interest rates) helps to increase the level of aggregate demand in the economy from AD_1 to AD_2, ceteris paribus. In theory, this increases real national income from Y_e to Y_f, thereby removing the recessionary gap, i.e. the difference between equilibrium national output (Y_e) and the national output at full employment (Y_f), with the general price level increasing from P_{L1} to P_{L2}.

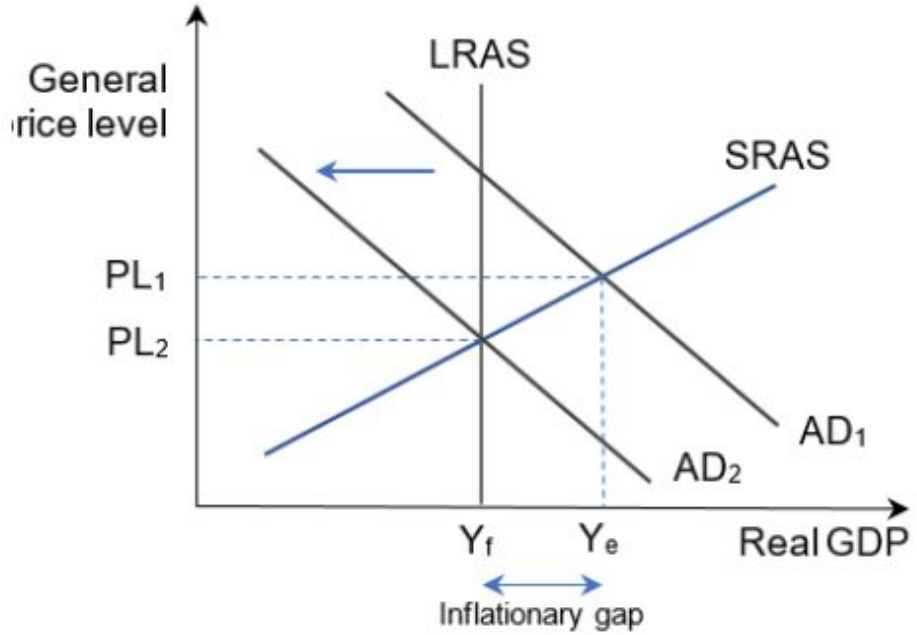
Award *1–2 marks* for an answer that shows a correct diagram OR an accurate written response.

Award *3–4 marks* for an answer that shows a correct diagram AND an accurate written response, similar to the example above.

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[Paper 2 and 3 Exam Practice Question 24.4]



A contractionary monetary policy (by raising interest rates) shifts the aggregate demand curve leftwards from AD_1 to AD_2, ceteris paribus. In theory, this reduces real national income (real GDP) from Y_e to Y_f, thereby eliminating the inflationary gap. This means that actual real national output is brought in line with the long-run aggregate supply (potential output) at Yf, with the general price level returning from P_{L1} to P_{L2}.

Award *1–2 marks* for an answer that shows a correct diagram OR an accurate written response.

Award *3–4 marks* for an answer that shows a correct diagram AND an accurate written response, similar to the example above.

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Constraints of monetary policy

Limited scope of reducing interest rates, when close to zero

Low interest rates do not necessarily **pull an economy out of recession**.

For example, **central banks cut interest rates** close to **0%** during the **COVID-19 pandemic**.

This means that **interest rates cannot be lowered further** to stimulate the economy.

Consumer and firms **might not respond** to lower rates if **confidence is low.**

Expansionary monetary policy is **not necessarily effective.**



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Real world example

Video: What 0% Interest Rate Really Means

How do zero interest rates discourage saving?

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Constraints of monetary policy

Low consumer and business confidence

During the **2008 financial crisis,** even though **interest rates were nearly zero**, the lack of **consumer and business confidence** led to a **prolonged recession.**

Changes in interest rates and money supply can be destabilizing to firms and consumers.

At times of **recessions**, consumers may be **unwilling to purchase goods and services**.

Firms may **not borrow money** to invest even though interest rates are low.

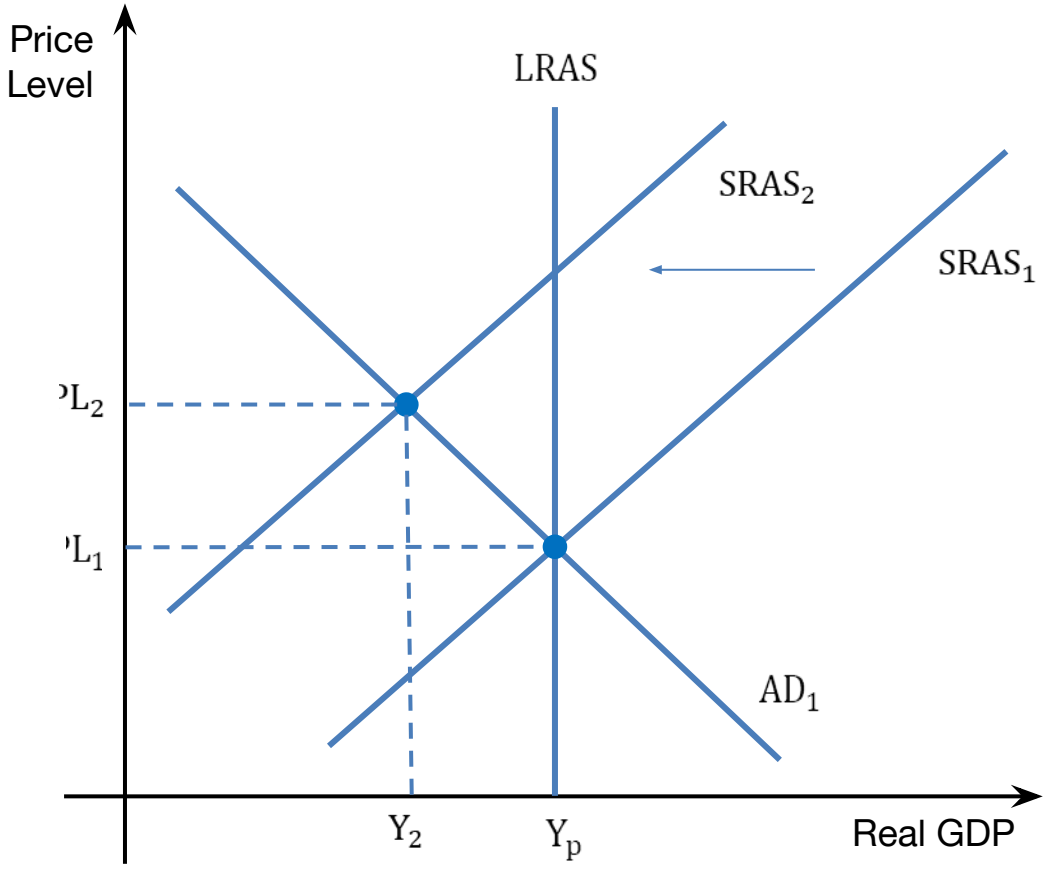
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Constraints of monetary policy

Trade-offs with other macroeconomic aims

A key limitation of contractionary demand side policy comes with the **trade-offs** when dealing with **cost-push inflation**.

When the economy experiences **cost-push inflation**, **price levels increase** from PL1 to PL2 while **real GDP decreases** to Yp to Y2.



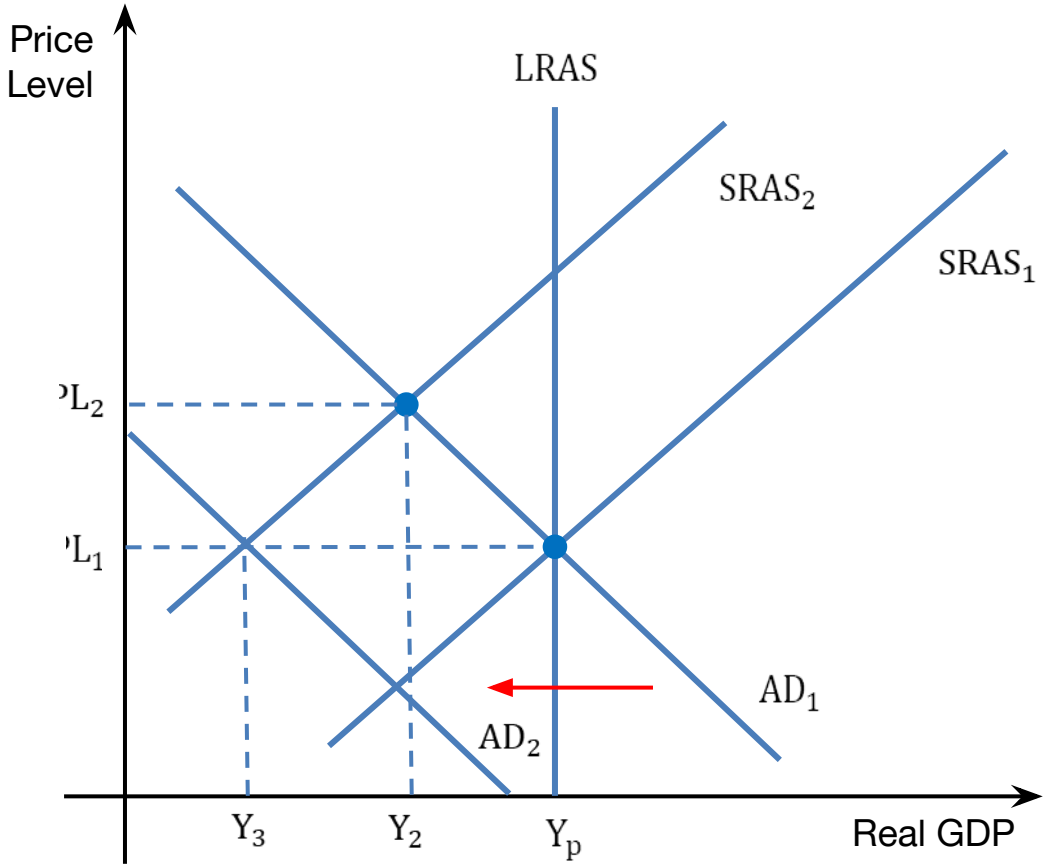
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Constraints of monetary policy

Trade-offs with other macroeconomic aims

In order to maintain a **low and stable rate of inflation**, **contractionary monetary** (or fiscal) **policy** may be used to reduce AD1 to AD2.

While this may **reduce the rate of inflation** back to a low and stable rate at PL1, there is a **trade-off** with two other **macroeconomic objectives** of **economic growth** and **unemployment** as **real GDP decreases further** to Y3 resulting in a **greater recessionary gap.**



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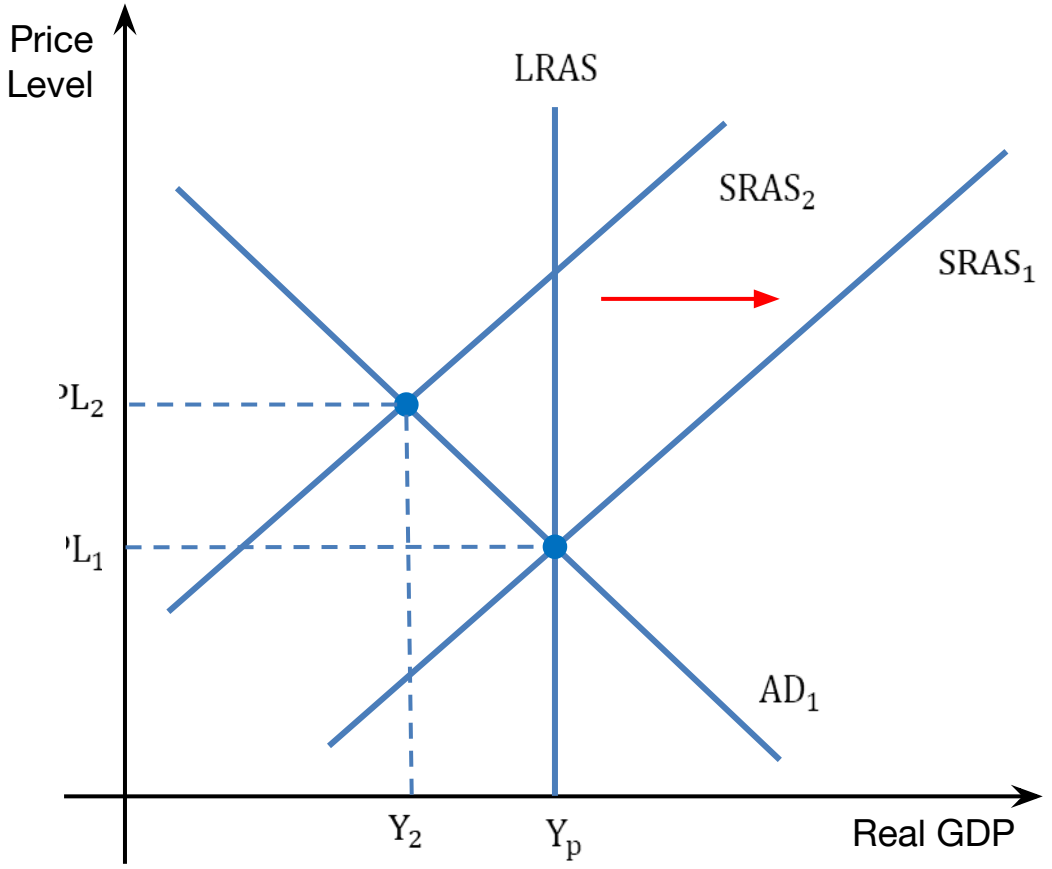
Constraints of monetary policy

Trade-offs with other macroeconomic aims

Therefore, when addressing **cost-push inflation** in an economy, it is more suitable to implement **supply-side policies.**

This addresses the objectives of **low and stable inflation**, **economic growth**, and **low unemployment**.

However, supply-side policies may have considerable **time-lags.**



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Strengths of monetary policy

Incremental, flexible and easily reversible

Incremental – Interest rates can be adjusted incrementally (at 0.25%) to reduce the risks of causing huge disruptions in the economy.

Flexible – Since the central bank is independent from political interferences, it has the flexibility to act at the best interest of the economy.

Reversible – The decision is reversible. For example, if the money multiplier is underestimated and thus causing inflation, the central bank can increase the interest rates immediately.

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Strengths of monetary policy

Possibly **shorter time lags** (compared to fiscal and supply-side policies)

Monetary policy can be **implemented more quickly** compared to **fiscal policy**.

For example, the **Bank of England** cut interest rates **twice in one week** in **March 2020** to stabilize the economic downturn.



However, a certain degree of **time lag** is still inevitable as time is needed for households and firms to **react to changes in interest rates**.

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Research activity (groups of 3-4)

1. Select a central bank of your choice
2. Research the central bank's monetary policy conducted over the past 5 years.
3. How effective has the central bank been in achieving their aims?

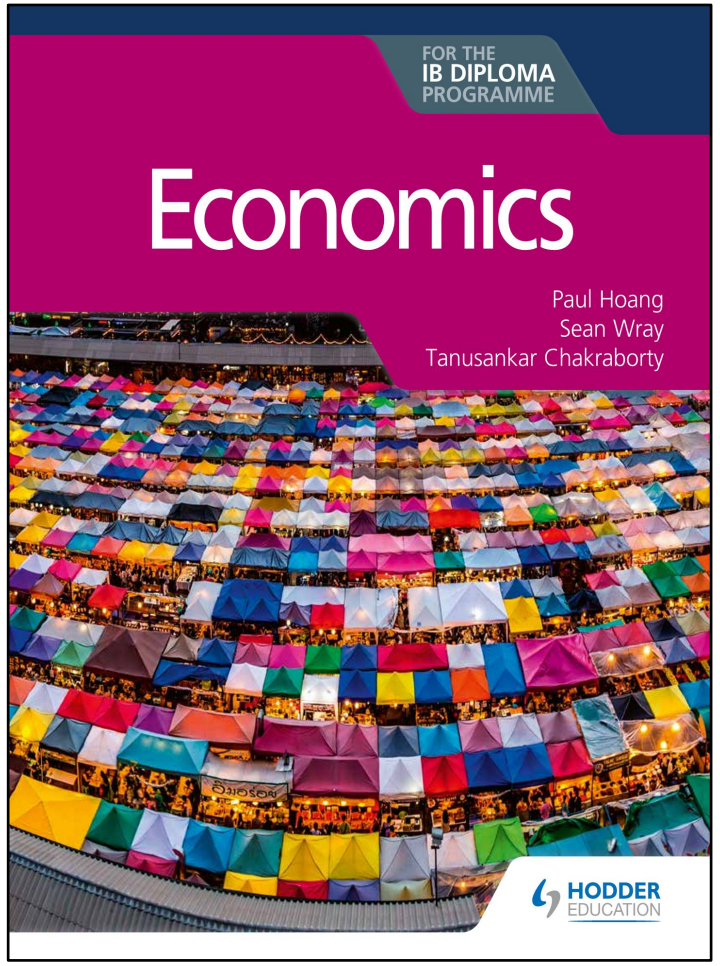
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- Page 378-379
- Paper 1 Exam Practice Question 24.5
- [10 marks]
- Paper 1 Exam Practice Question 24.6
- [15 marks]



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PAPER 1 EXAM PRACTICE QUESTION 24.5

Explain how monetary policy can be used to influence the level of economic activity. *[10 marks]*

PAPER 1 EXAM PRACTICE QUESTION 24.6

Evaluate the effectiveness of monetary policy in achieving macroeconomic objectives. *[15 marks]*

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[Paper 1 Exam Practice Question 24.5]

Use the assessment criteria on page 1. Award up to the maximum marks as indicated. Answers may include:

- Definitions: monetary policy and economic activity.
- Explanation: that increasing interest rates, as part of tight monetary policy, can be used to contract economic activity if there are inflationary pressures (vice versa). Explanation that currency appreciation can deflate the economy during times of rapid economic boom and overspending on exports (vice versa). Explanation that increasing the money supply during times of economic downturn can help to improve liquidity in the economy (although this may have inflationary effects). Explanation that lower interest rates also tend to lead to a lower exchange rate, thus causing an increase in the demand for the country's exports. Explanation that monetary policy is used to control inflation in order to achieve macroeconomic objectives. For example, reducing interest rates to stimulate economic activity means that:
    - Households and firms with outstanding mortgages have lower interest payments to make, thus have more disposable income to spend, i.e. consumption increases.
    - Savers have less of an incentive to deposit their money in banks, so are more likely to spend their money. This raises the level of aggregate demand in the economy, ceteris paribus.
- Diagram: appropriate use of an AD–AS diagram(s) to show the effects of the above explanations.

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[Paper 1 Exam Practice Question 24.6]

Use the assessment criteria on page 2. Award up to the maximum marks as indicated. Answers may include:

- Definitions: monetary policy and macroeconomic objectives.
- Explanation: of the use of monetary policy to achieve economic growth, low unemployment, as well as low and stable inflation. Explanation that interest rates can be used to influence the level of consumption, investment, government expenditure (and government debt), and net exports. Explanation that time lags between changes in the interest rate and money supply and their impacts on the economy can be destabilising. Explanation that restricting monetary growth by increasing bank reserve requirements can cause government (rather than market) failure, thereby causing banks to become less efficient and less productive. Explanation that inflationary pressures can cause a price-wage spiral in the labour market, thereby limiting the effectiveness of any tight/contractionary monetary policy.
- Diagram: AD–AS diagram(s) to show the economic impacts of changes in interest rates in the economy.
- Synthesis (evaluate): strengths and limitations of using monetary policies to achieve macroeconomic objectives. Consideration of the limited success in affecting aggregate demand if the economy is in an economic depression and/or if consumer and business confidence levels are low. Consideration of potential conflict between macroeconomic objectives, such as lowering interest to stimulate the economy which could fuel inflation in the economy. Acknowledgement that aggregate demand (and consumption and investment expenditure in particular) are not totally and solely dependent on interest rates.
- Examples: real-world examples of countries that have experienced success and/or challenges in using monetary policies to influence the level of economic activity, such as during the global financial crisis of 2008 and/or the global coronavirus pandemic of 2019/2020.

Note: It should be noted that definitions, theory and examples that have already been given in part (a), and then referred to in part (b), should be rewarded.

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Test your knowledge on this unit: Kahoot!

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